

How does the open chromatin landscape differ between a differentiated fibroblast-like cell state and a stem-cell-like iPSC state?

An introduction to Bulk ATAC-seq data analysis

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■ Suppose you want to know – “Which genes are active inside a cell?”

You might think we simply measure RNA. That is what RNA-seq does. But there is an earlier step. Before a gene can produce RNA:

1. DNA must be physically assessable.
2. Regulatory proteins must bind RNA.
3. Transcription begins.

ATAC-seq measures step 1. It tells us, which parts of the genome are opened and available for transcription.

DNA does not sit naked in the nucleus. It wraps around protein complexes called nucleosomes. Each nucleosome is an octamer — a group of eight histone proteins (two copies each of H2A, H2B, H3, and H4) — and approximately 147 base pairs of DNA wind around it about 1.7 times. The resulting structure looks like a bead on a string: the bead is the nucleosome; the string is the linker DNA connecting adjacent beads. (Figure 1)

Think like a statistician:

Imagine a library. The genome is one enormous library. Every cell processes exactly the same library. The question is:

Which books are currently open? Not, Which books exist?

Every cell contains-

- ∞ BRCA1- *A tumour suppressor gene that produces a protein to repair damaged DNA.*
- ∞ TP53- *A famous tumour suppressor gene often called the “guardian of the genome”.*
- ∞ INS- *Specific gene, responsible for giving instruction to make the hormone insulin.*
- ∞ Albumin- *Blood protein.*

All genes are present. But a liver cell reads different books (genes) than a neuron.

RNA-seq: Which books are actually being copied.

■ Why DNA must be packed ?

2 meters of DNA must fit into a nucleus only 6 μm wide. This is like fitting 2km of thread into a tennis ball. Cells solve these using proteins called histones. DNA wrapped around histones. The DNA-histone complex is called chromatin.

■ Nucleosomes ?

The DNA wraps around an octamer of histones; a group of eight **histone proteins** (2 copies of H2A, H2B, H3, H4) (Figure 2). Think simply, Suppose DNA is a very long string. Every 147 base pairs, - the string wraps around a spool. That spool is a nucleosome.

Open V/s Closed Chromatin:

This is the most important biological idea.

Open Chromatin: It is also called a **Euchromatin**. Large gaps exist. Proteins can bind. Genes can be activated.

Closed Chromatin: It is also called a **Heterochromatin**. Everything is compressed. Proteins cannot reach DNA. Genes remain silent.

Figure 1: Chromatin Accessibility

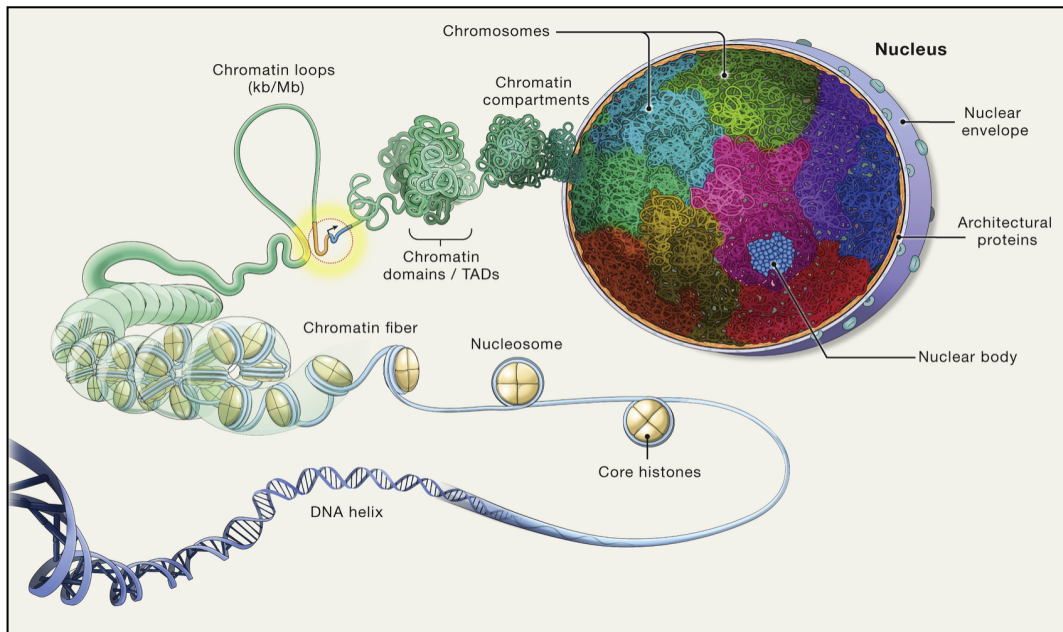


Figure 2: Histone protein/ Nucleosome

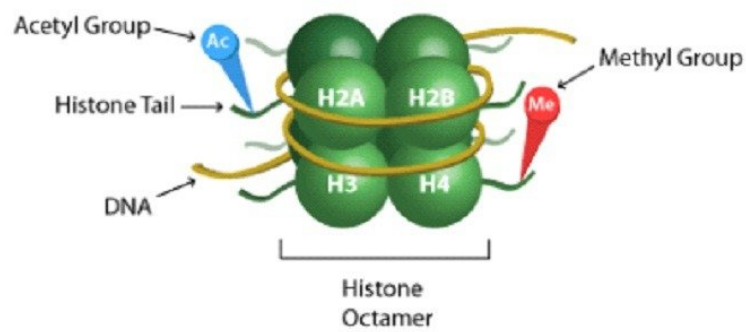
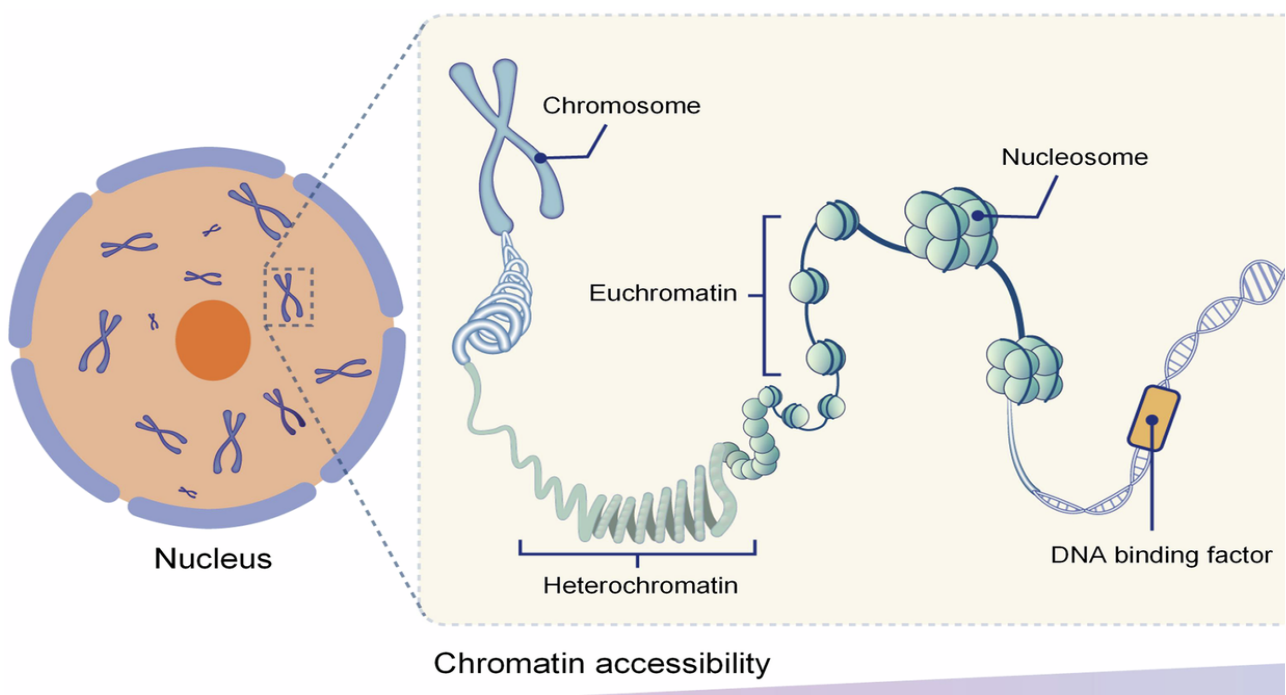


Figure 3: Euchromatin & Heterochromatin



▪ Accessibility is a Latent Variable

As statisticians we often measure latent variables. Examples: Depression, SES, Quality of Life, Cardiometabolic Risk etc.

Accessibility is another latent variable. We cannot directly observe “how open DNA is” instead, ATAC-seq measures evidence of openness.

▪ Why Different Cells Behave Differently?

Every cell has the same genome. But different cells open different regions. For example:

Neuron	Liver
Gene A: Open	Gene A: Closed
Gene B: Closed	Gene B: Open
Gene C: Open	Gene C: Closed

Therefore,

- ∞ Different genes are expressed.
- ∞ Different proteins are produced.
- ∞ Different cell types emerge.

This explains why muscle, neurons, blood cells all have identical DNA yet completely different functions.

▪ What does ATAC mean?

ATAC-seq stands for **Assay for Transposase-Accessible Chromatin using sequencing**. It was published in 2013 by Buenrostro and colleagues at Stanford and has since become one of the most widely used methods in epigenomics. Its power lies in its simplicity.

Key idea→ Use an enzyme that inserts only where DNA is exposed.

▪ The Tn5 Enzyme?

Think of Tn5 as a survey interviewer. Suppose houses represent DNA.

Closed houses→Door locked→Interviewer can't enter.

Open houses→Door open→Interviewer enters and leaves a sticker.

This sticker becomes the sequencing adapter. Thus, more stickers⇒More sequencing reads⇒Higher accessibility.

Scientifically, the Tn5 transposase enzyme structure is a dimeric molecular assembly where two protein subunits work together to bind double stranded DNA molecules. The Tn5 transposase is a 476-amino acid enzyme that functions as a homodimer to catalyze the movement of Tn5 transposons. It belongs to the DDE/D superfamily of polynucleotidyl transferases. Looking at the Figure 4, you can see this symmetrical arrangement formed by two interconnected protein units. The left side of the image displays one protein monomer winding into tightly coiled helices and left sheets, while the right side shows the second matching monomer interacting with it. Notice how two distinct segments of a DNA double helix are captured in this structure. One segment of DNA is positioned at the top right, while the second piece of DNA loops down at the bottom left. Each double stranded DNA molecule is held by both protein subunits, which actively orients the transposon ends directly into the active catalytic sites of the enzyme. This specialized structural arrangement gives the enzyme the precise geometric support needed to accurately cut and paste jumping genes across the genome.

▪ Entire ATAC-seq Experiment?

The process is surprisingly simple:

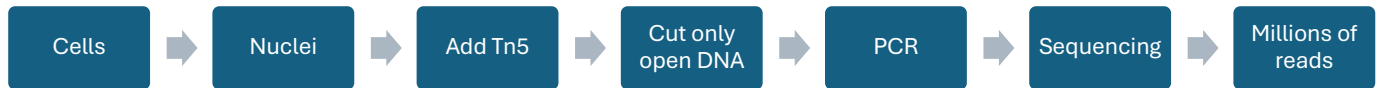
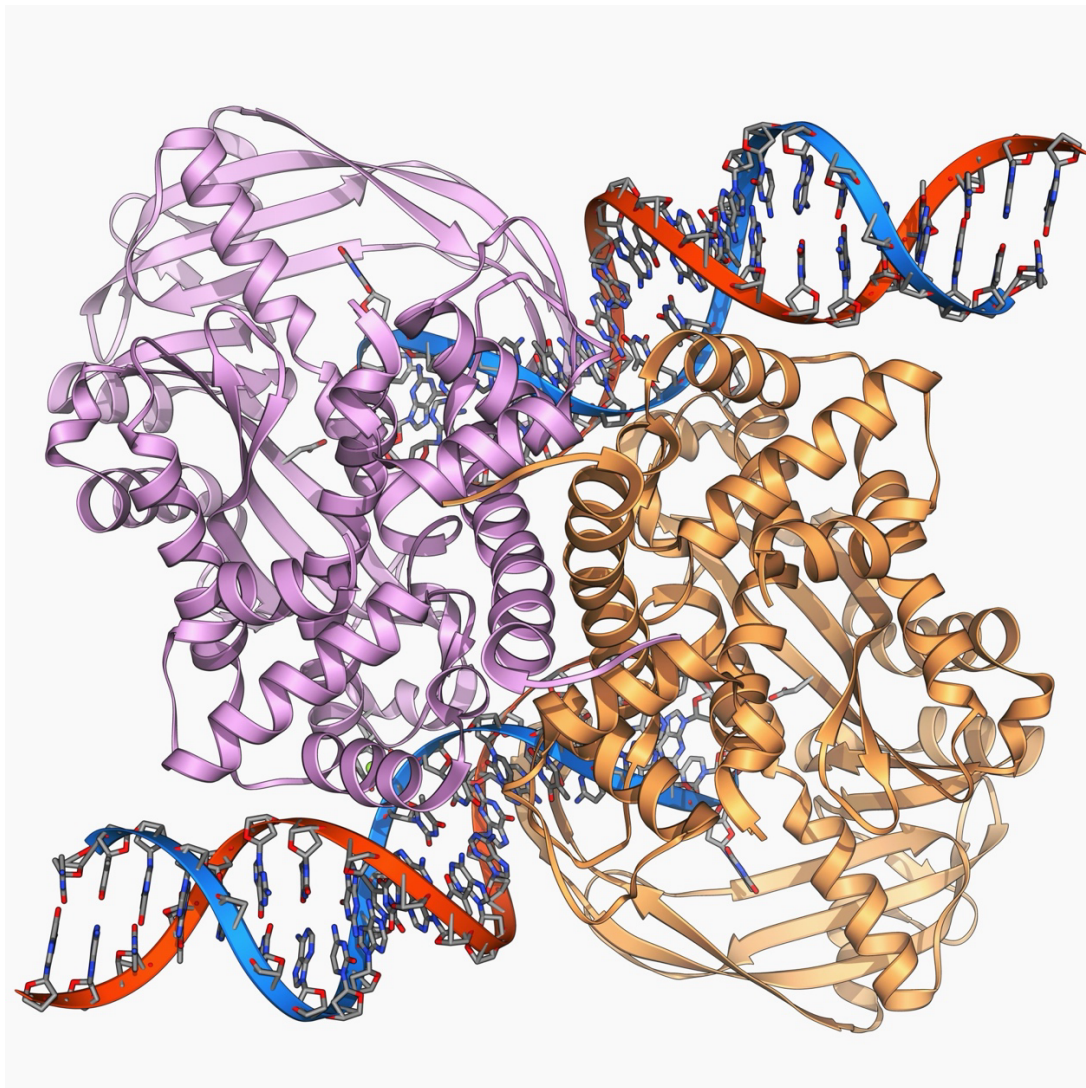


Figure 4: Tn5 transposase enzyme



▪ What is Actually Measured?

Suppose→ Promoter of gene X→ gets 10 reads

Suppose→ Promoter of gene Y→ gets 500 reads

Interpretation→ Gene Y promoter is much more accessible than X. Not necessarily more expressed. Only more open.

▪ What are Promoters?

First understand what is a Gene? A gene is a segment of DNA that contains the instructions for making a protein. Think of a gene as a recipe in a cookbook. However, a recipe cannot be used unless someone opens the page and starts reading it. The starting point is the promoter.

A promoter is a specific DNA sequence (with typically 100 to 1000 bp) located immediately before (upstream of) a gene transcription start sites (TSSs). Its job is to tell the cell: “Start reading the gene from here”. It is not the part of the protein-coding sequence. Instead, it is a regulatory DNA region. A simplified gene looks like this:

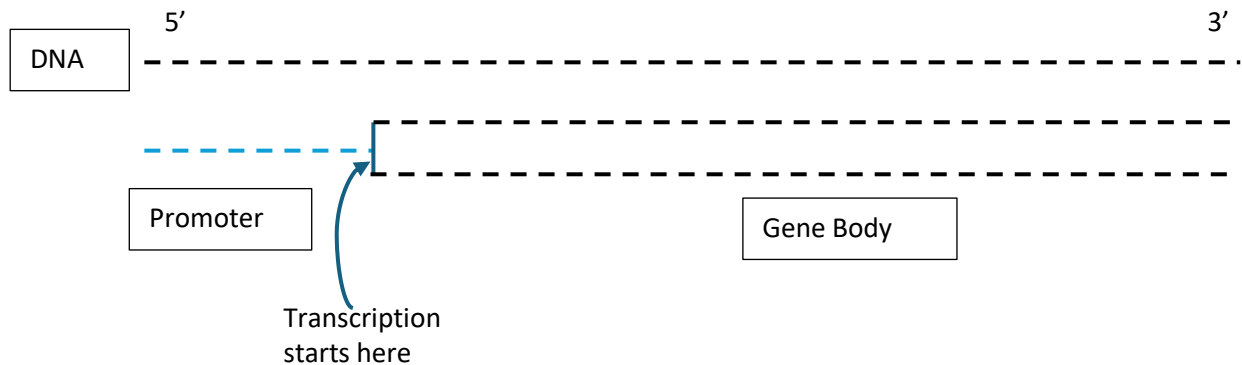
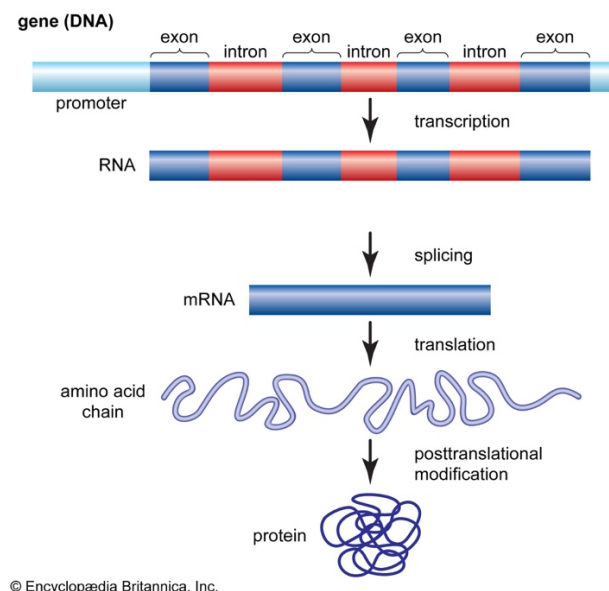


Figure 5: DNA to RNA transcription



What happens at the promoter?

The promoter is the place where proteins involved in transcription assemble. The main players are: (1) **Transcription Factor (TF) protein** and (2) **RNA Polymerase II**. The process is:



No promoter → RNA polymerase cannot efficiently start transcription (Figure 5).

Is every promoter active?

No, some promoters are open, accessible others are closed, inaccessible. This determines whether a gene can be transcribed.

Thus,

More reads → Higher accessibility

Fewer reads → Lower accessibility

Why is the promoter important in ATAC-seq?

ATAC-seq does not measure proteins or RNA. Instead, it asks a simple question: “Can the transcription machinery physically reach the promoter?” If the answer is yes, Tn5 inserts into the promoter, generating many sequencing reads. If the answer is no, very few reads are produced.

■ What are Transcription Factors (TFs)?

Transcription factors are specialized proteins that bind to DNA and regulate gene expression. They act as molecular switches that tell genes when to turn on or off, controlling how DNA is copied into RNA.

They help recruit or block RNA polymerase, the main enzyme responsible for making RNA from a DNA template.

Cell specialization: They ensure that different cells in our body – like skin cells v/s brain cells – produce the exact proteins they need to function.

■ What are Enhancers?

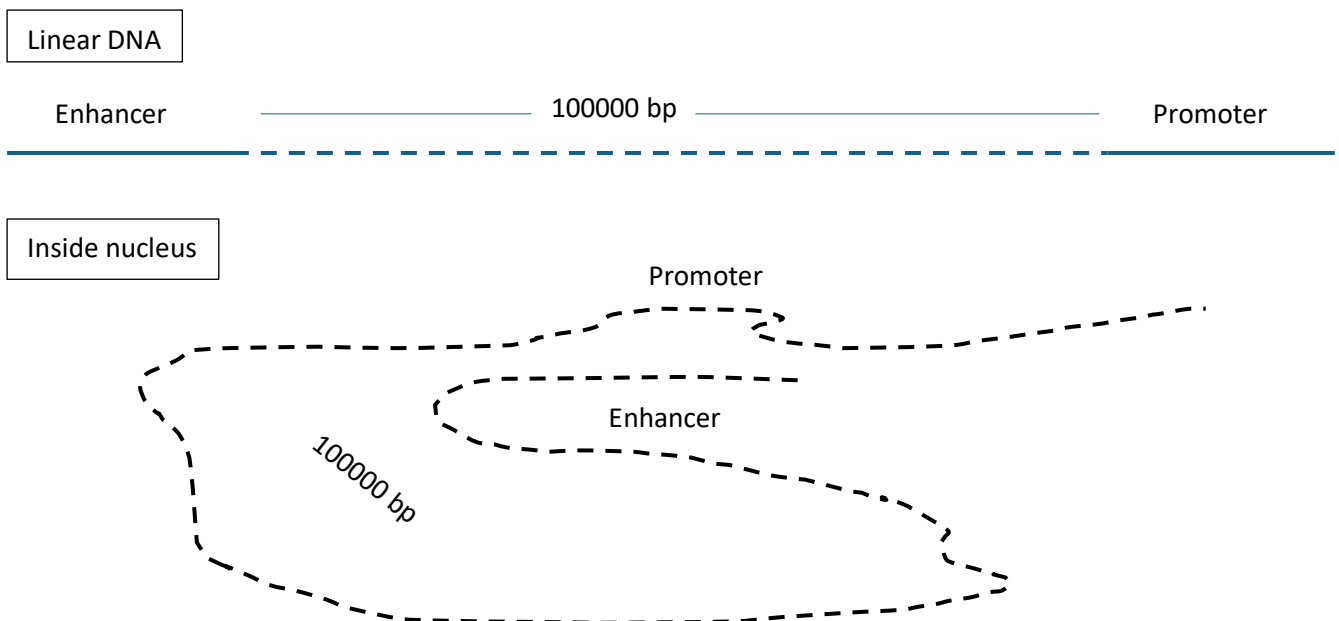
An enhancer is also a regulatory DNA sequence. However, its function is not to start transcription. Instead, it increases the efficiency or frequency of transcription. Think of an enhancer as the accelerator pedal in a car. The promoter starts the engine; the enhancer controls how fast the engine runs.

A formal definition would be: Distal regulatory elements, often thousands of base pairs away from their target genes, that bind clusters of transcription factors and physically contact promoters to boost gene expression.

Enhancers can be far away. Unlike promoters, enhancers do not need to be adjacent to the gene. They may be 1000 bp away, 100,000 bp away, even more than 1 million bp away. Surprisingly, the enhancer can still regulate the gene.

How can an Enhancer act from so far away?

DNA is not stretched out like a straight line. Inside the nucleus, DNA folds into loops. The loop brings the enhancer physically close to the promoter.



What binds to an Enhancer?

Enhancers are binding sites for specific transcription factors.



▪ What are Insulators?

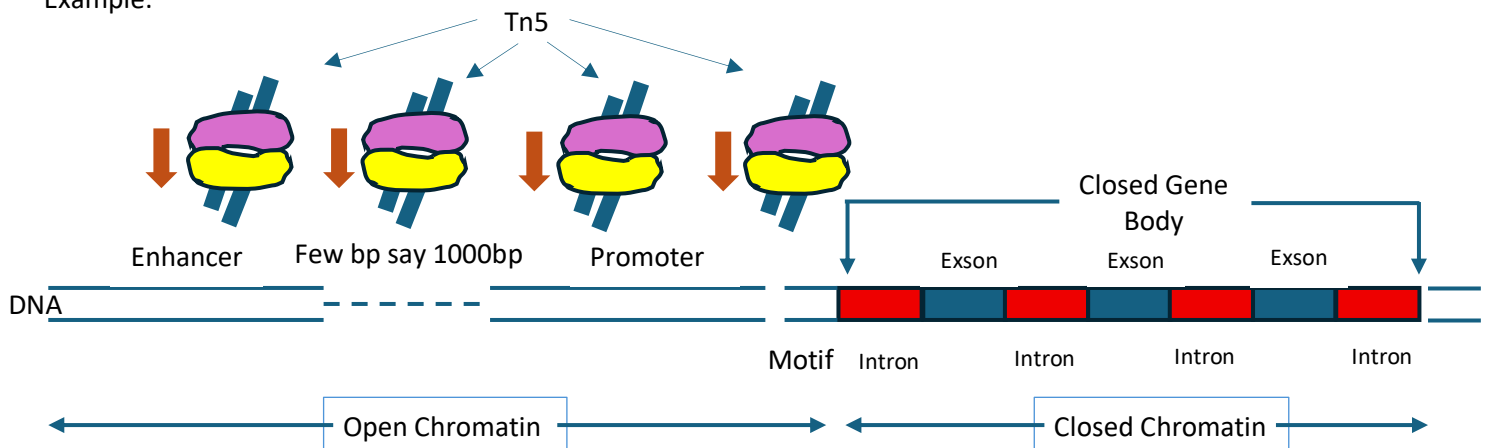
Boundary elements that prevent enhancers from activating the wrong genes.

▪ What are Transcription factor binding sites (Motifs)?

Short DNA sequences (motifs, typically 6–20 bp) that specific TFs recognize. Because transcription factors both require and create open chromatin, the ATAC-seq accessibility landscape essentially reveals the active regulatory vocabulary of the cell.

ATAC-seq measures chromatin accessibility. Therefore, all motifs, promoters, enhancers can be detected if they are open.

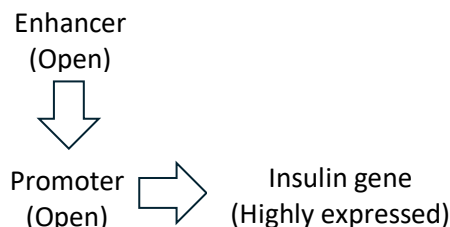
Example:



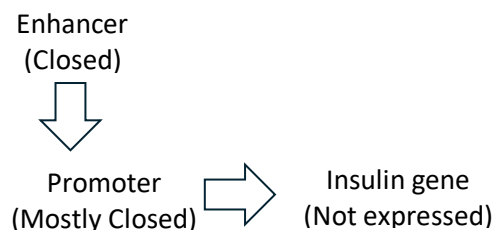
Tn5 inserts into both enhancer and promoter and the motif because they are accessible.

Example: Suppose we study the insulin gene.

In a pancreatic β -cell:



In a skin cell:



The DNA sequence is identical in both cells. The difference lies in chromatin accessibility and regulatory element activity.

■ Statistical Model for ATAC-seq?

Each genomic position has observed count say Y_i , where y_i = no. of Tn5 insertions.

If, $\uparrow y_i \Rightarrow$ means $\uparrow$ Accessibility. Therefore, ATAC-seq data are fundamentally count data across genomic coordinates. This is why downstream analysis often use Poisson, Negative Binomial or other count data models when biological replicates are available.

■ The ATAC-seq experimental protocol

1. **Cell harvest:** A small number of cells (typically 50,000–100,000) are collected.
2. **Nuclear isolation:** Cells are lysed with a mild detergent to release intact nuclei.
3. **Tagmentation:** Purified, hyperactive Tn5 transposase loaded with sequencing adapters is added directly to the nuclei for 30 minutes at 37°C. During this time, Tn5 inserts into accessible chromatin throughout the genome.
4. **DNA purification:** The reaction is stopped and DNA is purified.
5. **PCR amplification:** The tagged fragments are amplified. Because only accessible regions were tagged, only reads from accessible regions are amplified in significant numbers.
6. **Illumina sequencing:** The library is sequenced on an Illumina instrument, generating paired-end reads.

■ Fragment size distribution — a window into nucleosome positioning

One of the most informative quality metrics in ATAC-seq is the fragment size distribution — the histogram of how long each sequenced fragment is. This is not just a technical metric; it carries deep biological meaning.

- ∞ When Tn5 inserts twice in rapid succession into an open region, the resulting fragment is very short — less than 100 base pairs.
- ∞ These are called **sub-nucleosomal fragments** and they come from **nucleosome-free regions (NFRs)**, which are the most accessible stretches of the genome, typically found at active promoters and enhancer elements.
- ∞ When Tn5 cuts on either side of a **single nucleosome**, the resulting fragment is approximately 200 base pairs — the ~147 bp of DNA wrapped around the nucleosome plus a small amount of linker DNA.
- ∞ These are called **mono-nucleosomal fragments**.
- ∞ When Tn5 cuts around two adjacent nucleosomes, you get a **di-nucleosomal** fragment of approximately 400 bp, and so on.
- ∞ This creates a characteristic ladder pattern in the fragment size distribution: peaks at ~100–150 bp (sub/mono-nucleosomal), ~200 bp (mono-nucleosomal), ~400 bp (di-nucleosomal), ~600 bp (tri-nucleosomal).
- ∞ The presence of this ladder is one of the clearest signs that an ATAC-seq library worked correctly.
- ∞ A library that shows only a diffuse smear with no periodicity suggests poor chromatin accessibility or a failed tagmentation reaction.

Figure 5: Comparison of the distributions of fragment size between IMR90 and GM23338

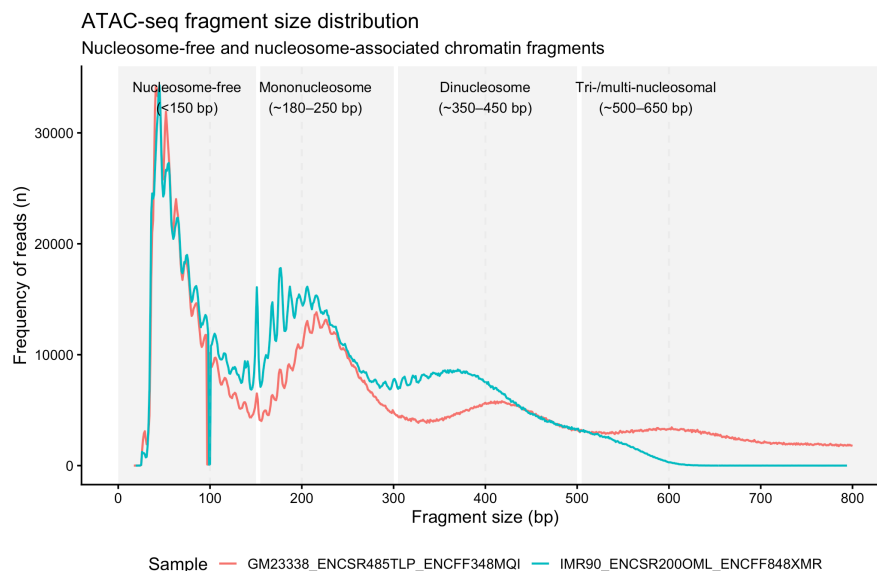
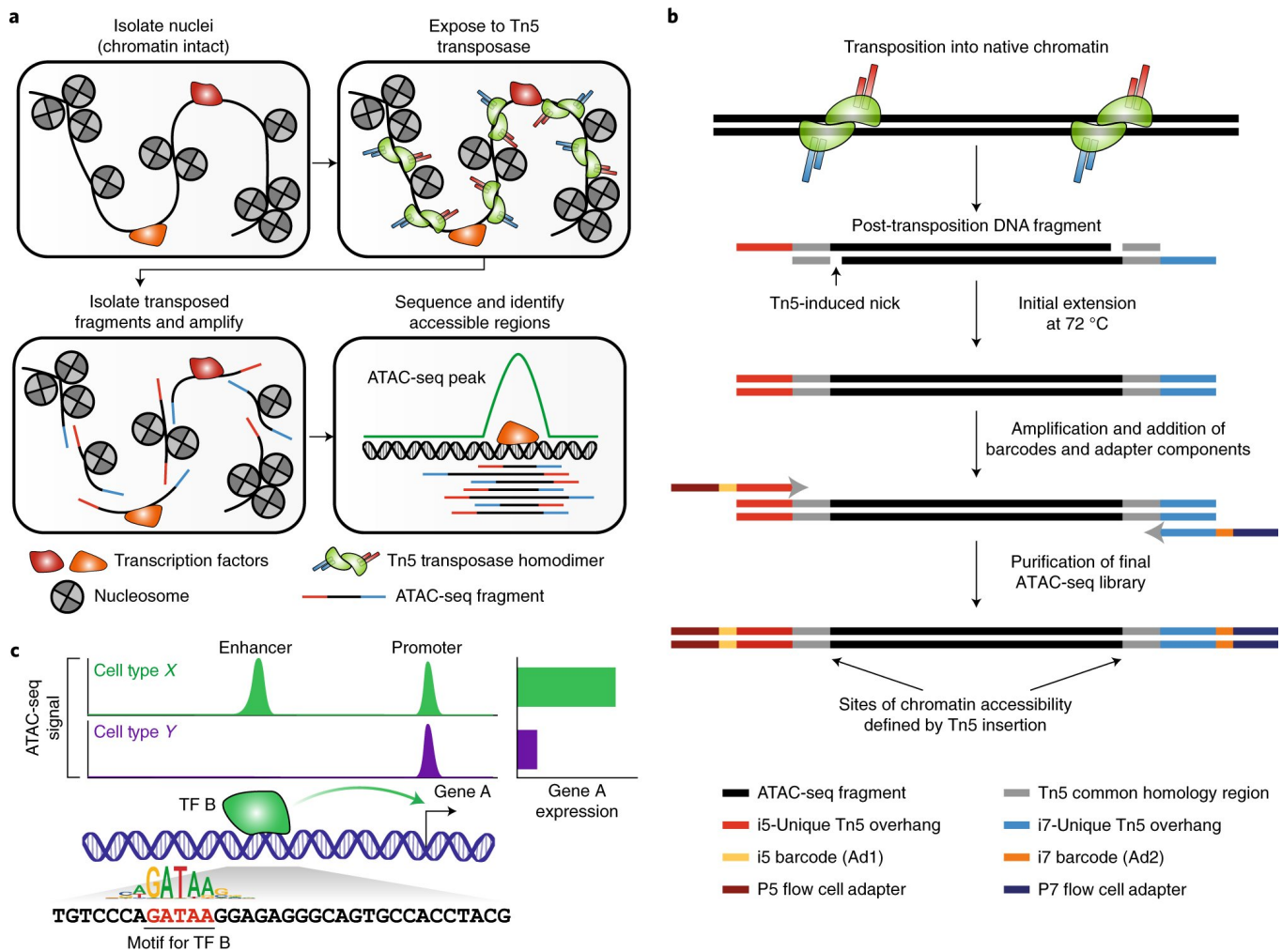


Figure 6: ATAC-seq Process and its Library preparation for sequencing.



■ Sequencing Library Preparation

Loaded Tn5 transposase reaches exposed DNA more easily and cuts the DNA into smaller pieces than nucleosome-wrapped DNA (Figure 6 (a)). To identify these reads, we need to sequence them in a Next-Generation Sequencer (such as an Illumina sequencer). But before that, we perform library preparation (Figure 6 (b)):

Step 1: Tagmentation (Cleavage & Adapter Ligation)

Process: The hyperactive Tn5 transposase simultaneously cleaves open chromatin regions and ligates short synthetic transposase adapter sequences to both ends of the DNA fragment.

Adapters Added:

- ∞ *Read 1 / Ad1-side adapter:* TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG
- ∞ *Read 2 / Ad2-side adapter:* GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG
- ∞ *Mosaic End (ME):* Both adapters share a core 19-bp mosaic-end region (AGATGTGTATAAGAGACAG) required for transposase recognition.

Step 2: Indexing PCR Amplification

Process: Polymerase Chain Reaction (PCR) is performed using index primers that anneal to the attached transposase adapters.

Functional Additions:

- ∞ *P5 & P7 Flow-Cell Sequences:* Attached to the outer ends to allow library fragments to bind to the Illumina flow cell for cluster generation.

- ∞ Sample Barcodes (i5 & i7 Indices): Unique index sequences are incorporated to allow multiplexing (pooling multiple samples into a single sequencing run).

Step 3: Library Purification & Size Selection

Process: Use AMPure XP magnetic beads (or column purification) to clean up the PCR products.

Objective: Remove unreacted primers, enzymes, and unwanted adapter-dimers (~120–140 bp) that can consume flow-cell capacity.

Step 4: Quality Control (QC) & Quantification

Quantification: Measure amplifiable library concentration using qPCR (or fluorometry/Qubit for total dsDNA).

Size Distribution: Analyze fragment length using an Agilent Bioanalyzer or Tapestation. A successful ATAC-seq library displays a characteristic nucleosomal periodicity pattern (multimodal peaks corresponding to nucleosome-free regions, mononucleosomes, dinucleosomes, etc.).

Molarity Calculation:

$$\text{Molarity}(nM) = \frac{\text{Concentration} \left(\frac{ng}{\mu L} \right) \times 10^6}{660 \times \text{mean fragment size (bp)}}$$

Step 5: Pooling & Sequencing Run Planning

Pooling: Denature and pool individual libraries by molar concentration to ensure equal read representation.

Read Budgeting: Aim for million aligned reads (25 million paired fragments) per human sample according to ENCODE benchmarks.

▪ Illumina Sequencing

DNA sequencing is simply the process of reading the exact order of the nucleotides (Adenine, Thymine, Cytosine and Guanine) across a strand of DNA. Knowing this sequence helps scientists diagnose genetic diseases, trace viral outbreaks, and customize cancer treatments.

In ATAC-seq, many sequencing techniques can be utilized like First generation — Sanger-seq, Next-Generation Sequencing (NGS) like Illumina-seq, SOLiD-seq, etc., but most common, cost effective and comparatively accurate (98%) sequencing — Illumina-seq is being used.

History:

First-Generation Sequencing (1977)

In 1977, scientist Frederick Sanger developed the first widely used method to sequence DNA. While accurate, "Sanger sequencing" was slow and expensive. Sequencing the first human genome using this old method took over 10 years and cost around \$3 billion.

The Pub Breakthrough (1997)

In 1997, two chemists at the University of Cambridge, Shankar Balasubramanian and David Klenerman, came up with a revolutionary concept over a pint of beer at a local pub. Instead of reading one piece of DNA at a time, what if you could read millions of DNA fragments simultaneously on a single glass slide using glowing colored lights?

Solexa to Illumina (1998–2007)

1998: The founders formed a startup called Solexa to commercialize the idea.

2004: Solexa acquired key "clustering" technology to boost the optical signal, making the process reliable.

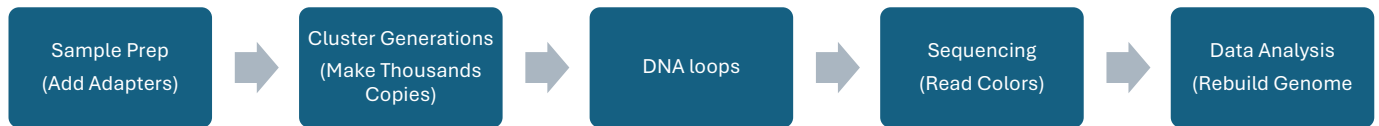
2006: Solexa released its first commercial sequencer, the Genome Analyzer.

2007: Biotechnology giant Illumina acquired Solexa for \$650 million.

Today, Illumina sequencing (also known as Next-Generation Sequencing, or NGS) powers over 80% of global DNA sequencing. It reduced the cost of reading a human genome from \$3 billion to under \$200, completing the process in just 24 hours.

Step-by-Step: How Illumina Sequencing Works

Think of Illumina sequencing like taking millions of microscopic, color-coded snapshots of growing DNA strands. The entire process follows **4 main steps**:



Step 1: Library Preparation (Prepping the DNA)

Already Explained in the previous section.

Briefly:

Raw DNA extracted from a cell is far too long to read all at once. It must be prepared into a "library" of short, readable pieces.

1. **Chopping the DNA:** The long DNA strand is chopped into short pieces, usually 200 to 500 bases long.
2. **Adding Adapters:** Synthetic pieces of DNA called adapters are glued to both ends of each fragment.
 - a. **Flow Cell Hooks:** Help the fragment stick to a special glass slide inside the machine.
 - b. **Barcodes (Indices):** Short unique code tags that identify which patient or sample the piece of DNA belongs to, allowing dozens of samples to be tested together.

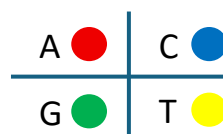
Step 2: Cluster Generation (Bridge Amplification)

A single microscopic DNA molecule is too faint for cameras to see. To fix this, the machine makes thousands of identical copies of each fragment in one spot, creating a cluster.

1. **Attaching to the Flow Cell:** The DNA fragments are poured onto a glass slide called a flow cell. The slide is coated with tiny lawn-like anchors that grab onto the adapters attached in Step 1.
2. **Forming a Bridge:** The attached DNA fragment bends over sideways to touch a neighboring anchor on the surface, forming a small arch or "bridge".
3. **Copying (Bridge Amplification):** Enzymes build a complementary copy of the bridged strand. The bridge unbends, releasing two single strands side-by-side.
4. **Repeating:** This bending, copying, and unbending cycle happens tens of times.
5. **The Result:** Around 1,000 identical copies of each original DNA fragment form a tiny, localized pile called a cluster. Millions of these distinct clusters sit side-by-side on the same flow cell.

Step 3: Sequencing by Synthesis (Reading the Letters)

Now the machine builds a new matching strand of DNA, letter-by-letter, while recording the process with high-powered cameras. Each DNA base is assigned a distinct fluorescent color tag:



Note: These exact color combinations vary by specific instrument optical model (e.g., 4-channel vs. 2-channel system).



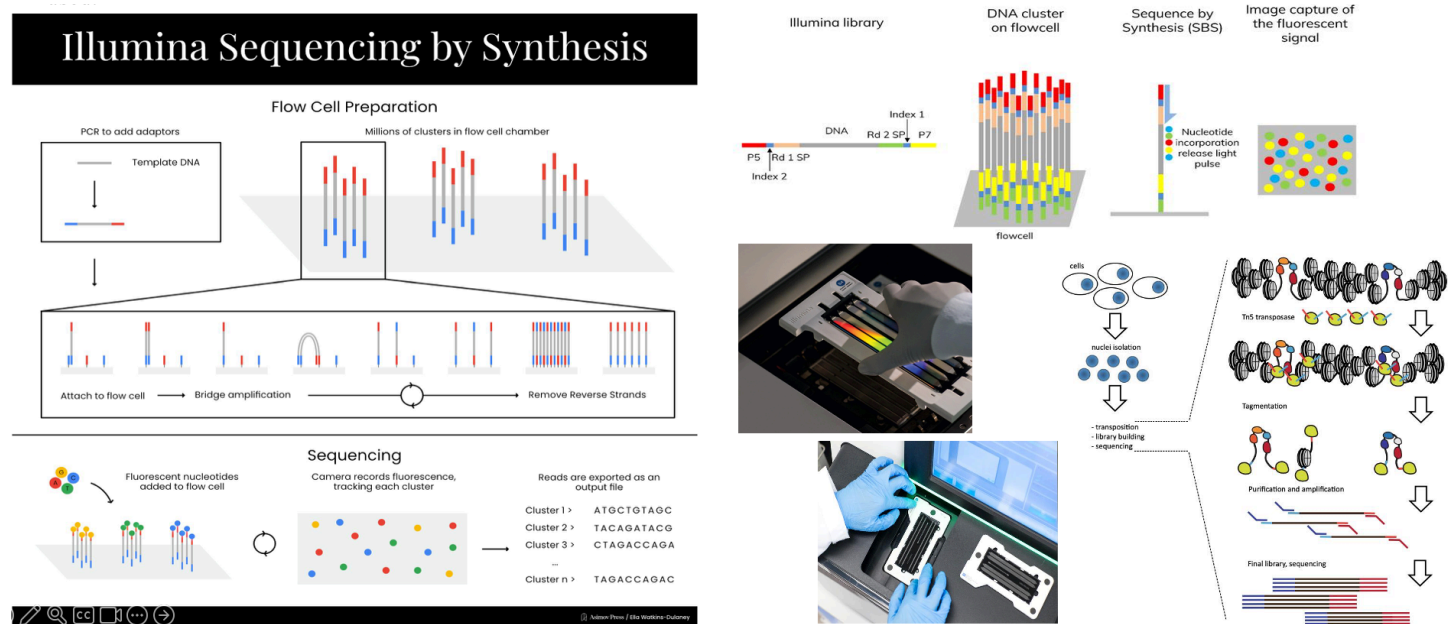
1. **Add One Base:** Fluorescently labeled building blocks (A, T, C, G) are flooded across the slide. They contain a chemical "brakes" lock that only allows one single letter to attach at a time.
2. **Snap a Photo:** High-powered lasers flashlight across the flow cell, exciting the color tag. The camera records the color emitted by each cluster. If a cluster glows blue, the next letter in that DNA sequence is a C.
3. **Unlock the Brakes:** A chemical wash removes the fluorescent dye and releases the chemical lock so the next base can attach.
4. **Repeat:** This cycle repeats 100 to 300 times, adding one letter per cycle and building the sequence line-by-line.

Step 4: Data Analysis (Putting the Puzzle Together)

Once the camera cycles finish, the machine turns millions of tiny colored images into raw genetic data.

1. **Base Calling:** Computer software converts the color sequence into text letters (A, T, C, G) along with a quality score (Q-score) indicating confidence in each letter read.
2. **Demultiplexing:** The computer reads the sample barcode tags (from Step 1) and sorts the reads into separate files for each patient or sample.
3. **Genome Alignment:** Bioinformatics software takes these short fragments (called "reads") and aligns them to a reference human genome map, much like using the box-art image to solve a massive jigsaw puzzle.

Figure 7: Steps in Illumina Sequencing.



What comes out: paired-end FASTQ files

Modern ATAC-seq always uses **paired-end** sequencing — the sequencer reads each fragment from both ends. This gives two reads per fragment:

Read 1 (R1): the forward read, starting from one end of the fragment

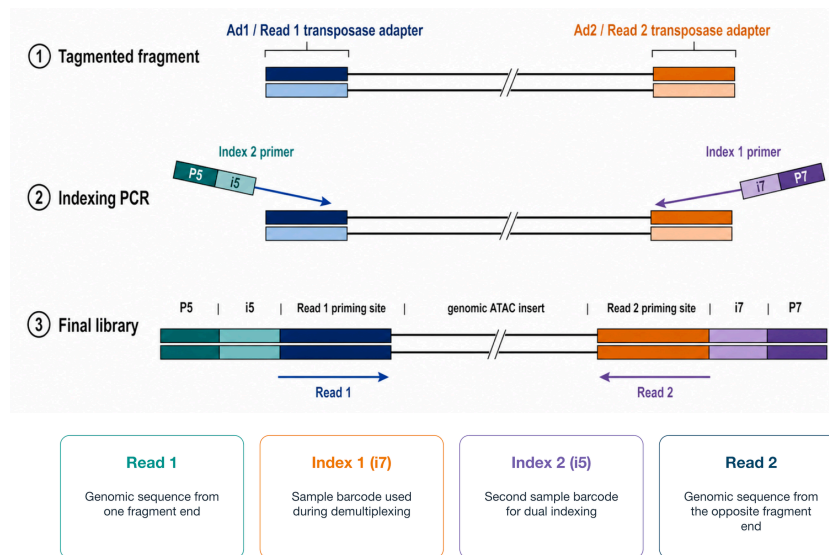
Read 2 (R2): the reverse read, starting from the other end

The raw output of the sequencer is a **FASTQ file** — a text file where each read is represented as four lines:

@SEQ_ID	← Read name/identifier
ACGTACGTACGT...	← DNA sequence
+	← Separator
IIIIHHHHGGG...	← Quality scores (Phred-encoded ASCII)

For a typical ATAC-seq experiment, two FASTQ files are produced per sample: one for R1 and one for R2.

Figure 8: Actual DNA segments and Read 1 and Read 2



■ From FASTQ to BAM — the pre-processing steps

In this tutorial, pre-aligned BAM files are provided — the analysis starts after the FASTQ-to-BAM conversion has already been completed. However, it is important to understand what happened to the raw reads to produce those files.

Step 1: Quality trimming (Trimmomatic / Trim Galore)

Raw FASTQ reads often contain adapter sequences from the sequencing library preparation — these are not genomic sequence and must be removed. Low-quality bases at the ends of reads ($Q < 20$) are also trimmed as they can cause misalignments.

Step 2: Alignment (Bowtie2 / BWA)

Trimmed reads are aligned to the human reference genome (hg38 — the Genome Reference Consortium Human Build 38). The aligner compares each read to the reference and assigns it to the most likely genomic location. The output is a SAM file (Sequence Alignment Map), which is then compressed to BAM (Binary Alignment Map) format.

Step 3: Post-alignment filtering

After alignment, several filtering steps are applied:

Remove PCR duplicates: PCR amplification during library preparation sometimes creates multiple copies of the same original DNA fragment. These must be removed because they do not represent independent transposase insertion events

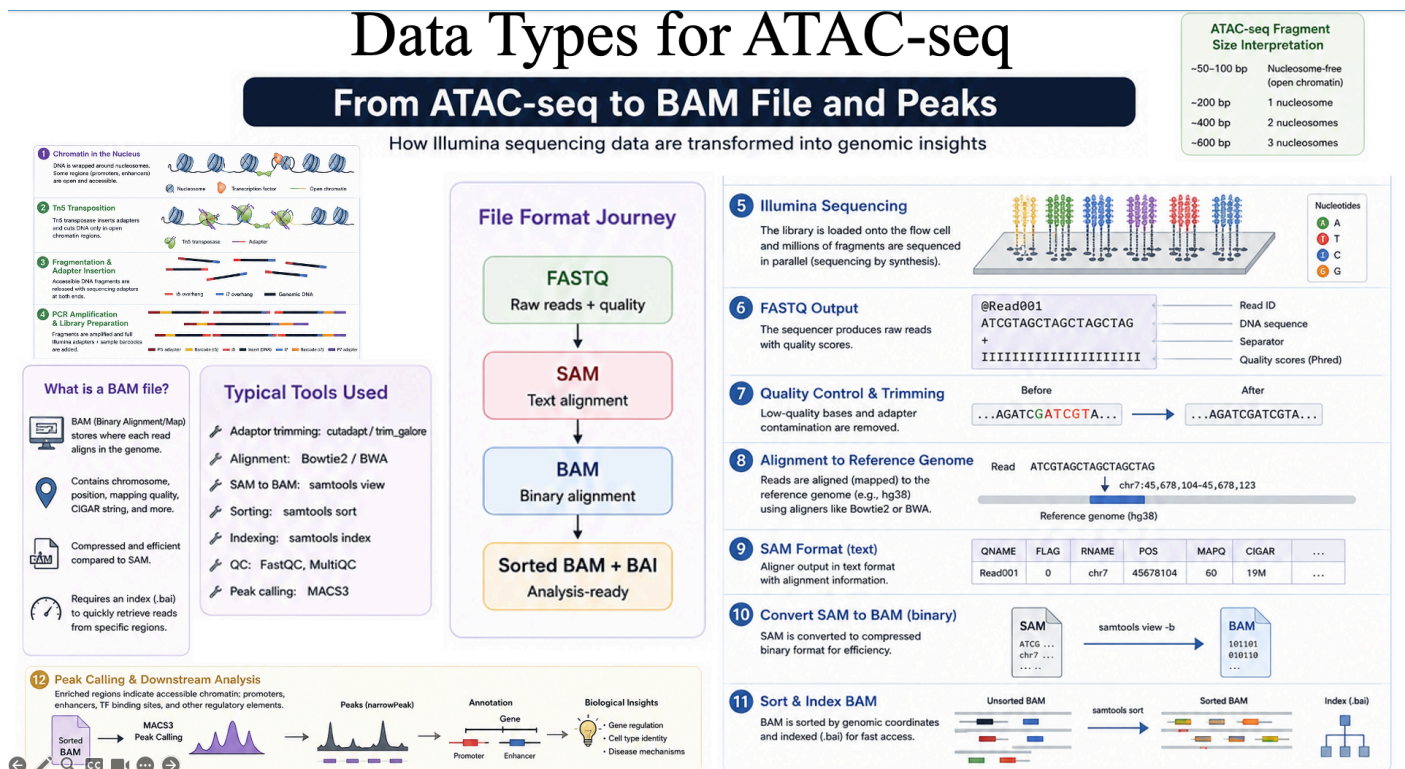
Remove mitochondrial reads: The mitochondrial genome is small, circular, and lacks nucleosomes. Tn5 cuts into it indiscriminately. Mitochondrial reads must be removed because they dilute the signal from the nuclear genome

Filter by mapping quality (MAPQ ≥ 30): Reads that aligned to multiple locations in the genome are filtered out. Only uniquely and confidently mapped reads are kept

Remove blacklist regions: Certain regions of the human genome (centromeres, telomeres, satellite repeats) give artificially high signal in all chromatin assays and are excluded using a published blacklist

The result of all this processing is the BAM file used in this workshop.

Figure 9: Complete file structure and data format used in ATAC-seq



■ The dataset — what are we comparing?

The two samples in this workshop represent a biologically meaningful contrast:

IMR90: is a cell line established from human fetal lung fibroblasts. It represents a differentiated, fibroblast-like cell state. In terms of chromatin, IMR90 cells have an accessibility landscape that reflects active mesenchymal/fibroblast gene programs — genes involved in extracellular matrix production, cell migration, and structural maintenance.

GM23338: is derived from an iPSC (induced pluripotent stem cell) line. iPSCs are reprogrammed from somatic cells back to a pluripotent state, resembling embryonic stem cells. Their chromatin is more open and permissive at pluripotency-associated gene loci — Oct4, Sox2, Nanog, and their downstream targets. GM23338 represents a stem-cell-like epigenomic state.

Both datasets were downloaded from the **ENCODE project** (Encyclopedia of DNA Elements, <https://www.encodeproject.org/>), a publicly funded, peer-reviewed repository of functional genomic experiments.

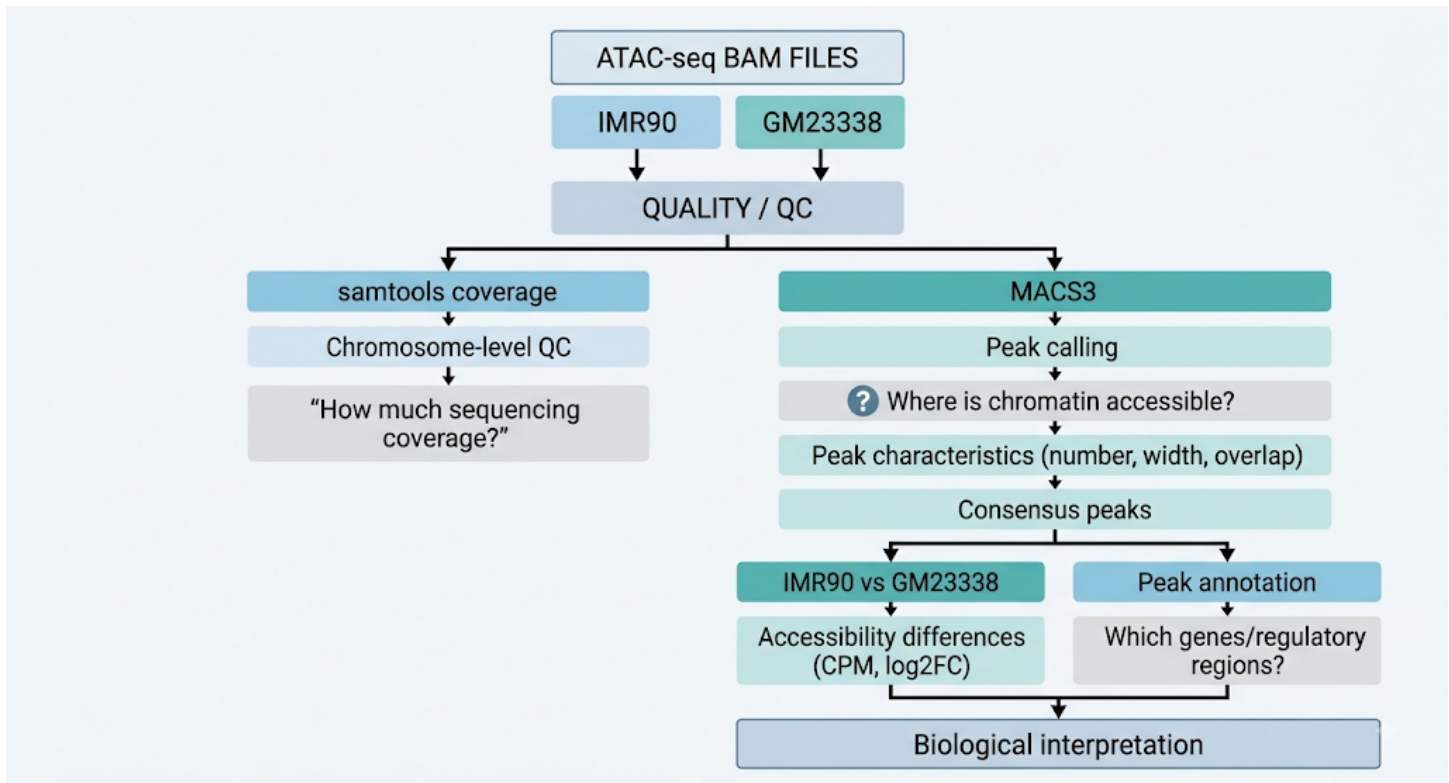
Sample	Cell line	ENCODE Experiment	ENCODE File
IMR90	Differentiated fibroblast-like	ENCSR200OML	ENCFF848XMR
GM23338	iPSC stem-cell-like	ENCSR485TLP	ENCFF348MQI

The biological question throughout this workshop:

How does the open chromatin landscape differ between a differentiated fibroblast-like cell state (IMR90) and a stem-cell-like iPSC state (GM23338)?

Important caveat: One BAM file is provided per cell type — no biological replicates. This means formal statistical differential accessibility testing cannot be performed. The analysis throughout this workshop is **exploratory**: we describe which regions appear more accessible in one cell type versus the other, without claiming statistical significance.

So, overall ATAC-seq Workflow is:

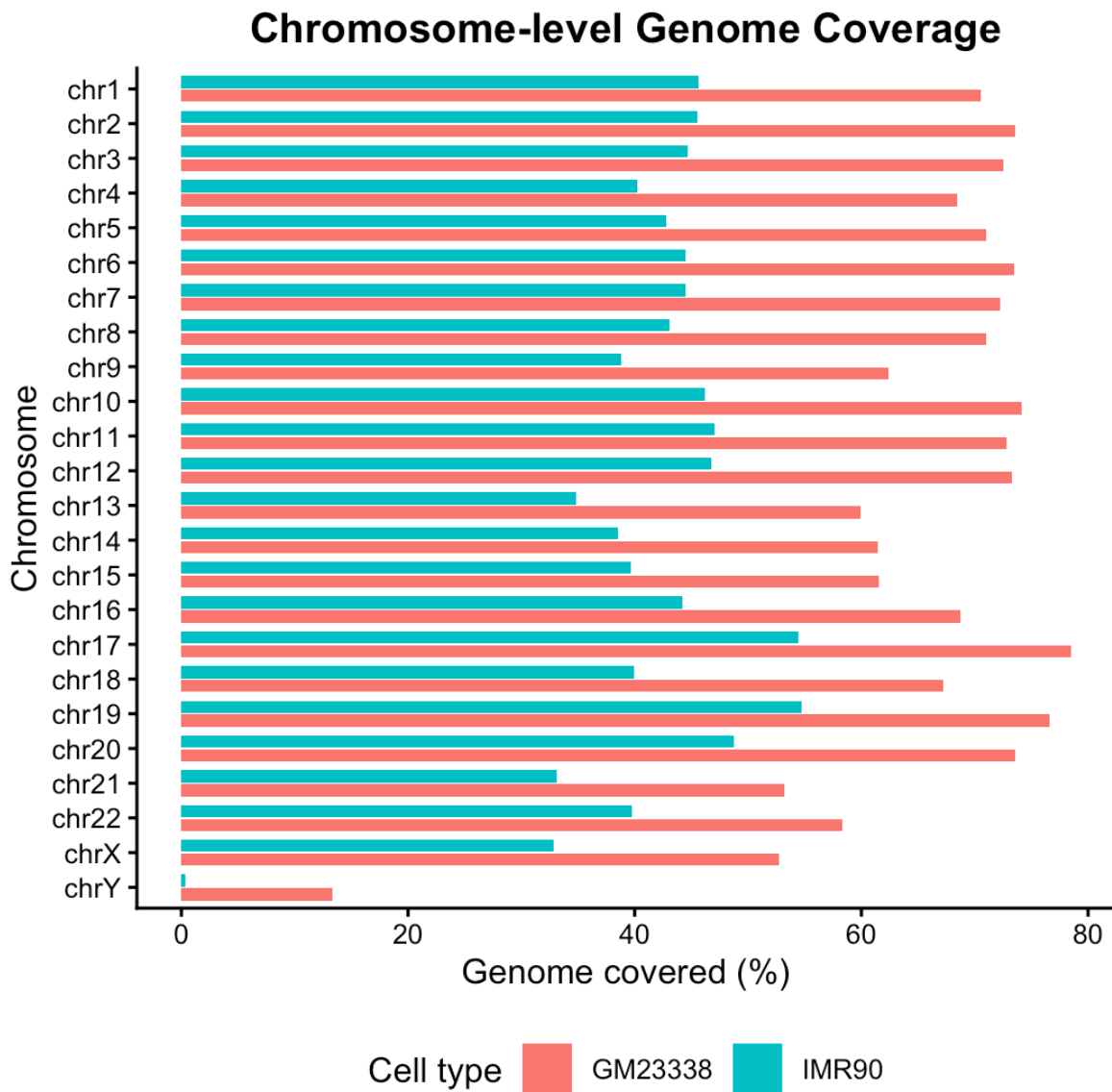


Analysis

■ Quality Checking of the Sequencing

- ∞ After removing the reads whose MAPQ was <30, removing PCR duplicates reads and filtering out the reads originating from mitochondrial DNA (which lacks histones and can contaminate ATAC-seq datasets), We found the total number of reads in IMR90 and GM23338 were 60310654 bp and 218889394 bp respectively.
- ∞ The mean MAPQ was observed 39.1 for IMR90 and 36.1 for GM23338.
- ∞ The average Coverage (percentage of chromosome bases covered by at least one aligned read.) was observed in GM23338 as 60-78% whereas for IMR90 approximately 35-55%.
- ∞ The highest coverage chromosome was Chr17 and Chr19: for IMR90: 11.15x and 13.70x respectively, whereas for GM23338: 3.15x and 3.86%
- ∞ **Overall QC interpretation:** The BAM files show **good sequence and mapping quality**, with GM23338 having considerably greater sequencing coverage/depth than IMR90.

Figure 10: Comparison of the distributions of fragment size between IMR90 and GM23338



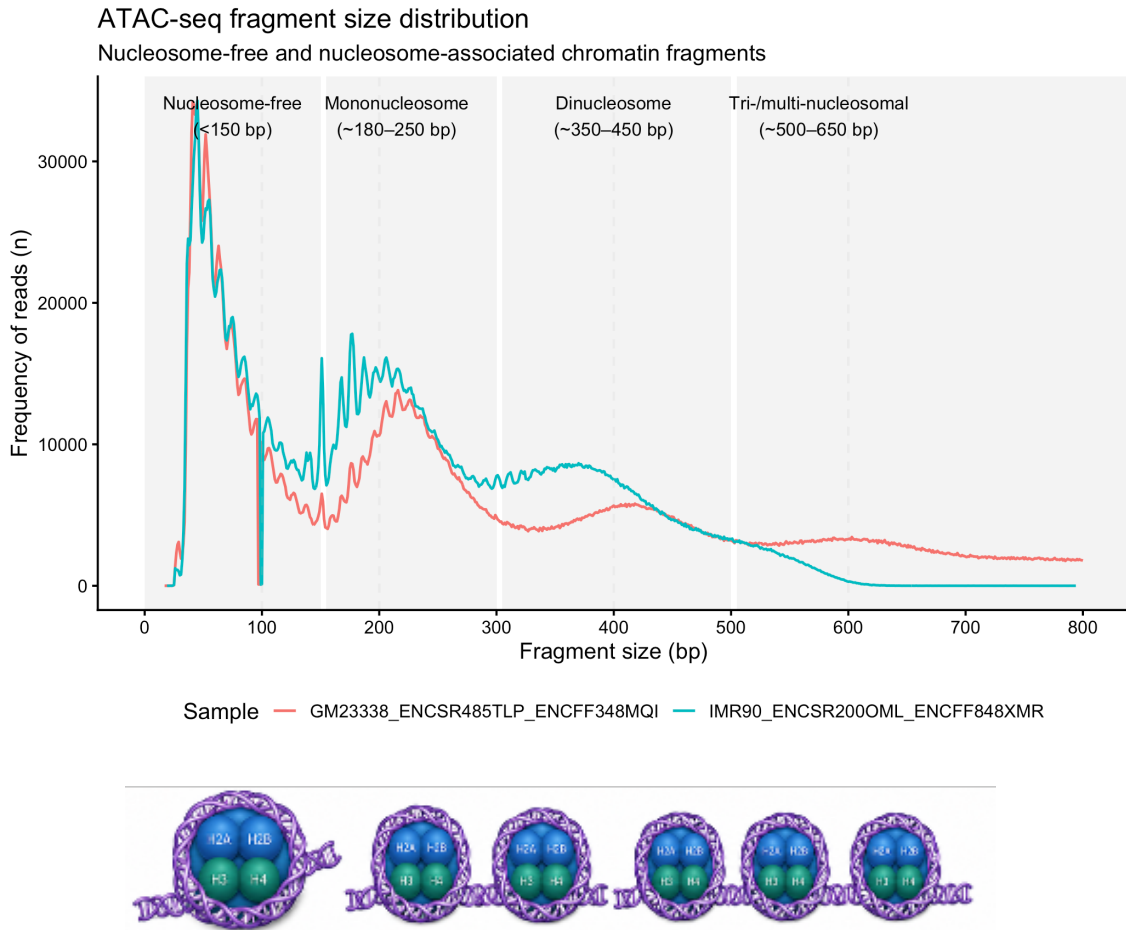
▪ Quality Checking of the Sequencing

A high-quality ATAC-seq dataset must satisfy two fundamental quality control (QC) criteria:

1. **Biophysical Integrity:** Preserved nucleosomal periodicity.
2. **High Signal Enriched over Background Noise:** A high Fraction of Reads in Peaks (FRiP > 20%).

The ATAC-seq fragment-size distribution assesses library quality and chromatin organization by **examining the relative abundance of nucleosome-free and nucleosome-associated DNA fragments (peaks)**; the expected enrichment near ~200, ~400, and ~600 bp reflects mono-, di-, and multi-nucleosomal fragment patterns.

Figure 11: Comparison of the distributions of fragment size between IMR90 and GM23338



Peaks are the specific genomic intervals (Chromosome: start - end) identified by a peak caller like **MACS3**. They mark regions where open chromatin allowed Tn5 transposase to cut, causing reads to stack up significantly above baseline genomic noise.

A read "**overlaps**" a peak if its alignment coordinates physically intersect or fall within the genomic boundary of any identified peak.

Looking at the fragment size distribution (Figure 11), we observe a classic **multimodal periodicity pattern** reflecting the physical wrapping of DNA around histones. The largest peak sits under **150 base pairs**, representing nucleosome-free regions where transcription factors bind. The secondary and tertiary peaks at **~200 and ~400 base pairs** correspond to mono- and di-nucleosomes. This confirms that Tn5 tagmentation was optimal and did not destroy chromatin architecture.

The **Fraction of Reads in Peaks (FRiP) score** is a core quality control metric in epigenomics (e.g., ATAC-seq, ChIP-seq, and CUT&RUN) that quantifies the signal-to-noise ratio of a sequencing experiment. It measures the proportion of clean, high-quality sequencing reads that map directly inside statistically significant signal clusters (peaks) relative to the genomic background noise.

Mathematical Definition

$$FRiP = \left(\frac{N_{peaks}}{N_{total}} \right) \times 100\%$$

- ∞ Numerator (N_{peaks}): The count of high-quality, non-duplicate, non-mitochondrial reads whose genomic coordinates physically overlap called peak regions (e.g., MACS3 peaks).
- ∞ Denominator (N_{total}): The total number of mapped, high-quality, deduplicated reads generated by the library.

Why FRiP Matters

- ∞ Precision & Efficiency: Functionally acts as a statistical precision metric, indicating how successfully the assay targeted open chromatin or factor-bound loci rather than generating uninformative background noise.
- ∞ Troubleshooting Indicator: A low FRiP score usually indicates experimental issues, such as poor enzyme/antibody efficiency, excessive dead cells, incomplete nuclear isolation, or over-transposition.

Standard QC Benchmarks

Assay Type	Target Signal	Minimum ENCODE Threshold	Optimal Quality Range
ATAC-seq	Open Chromatin / Enhancers	> 20%	30% - 60%
ChIP-seq (Point-source)	Transcription Factors (e.g., CTCF)	> 1%	> 5%
ChIP-seq (Broad-mark)	Histone Modifications (e.g., H3K4me3)	> 5%	> 10%

In our analysis Looking at the FRiP score, The Fraction of Reads in Peaks measures our signal-to-noise ratio. Both GM23338 at ~55% and IMR90 at ~52% far exceed this threshold, proving that over half of our sequencing data represents true, highly enriched biological signal.

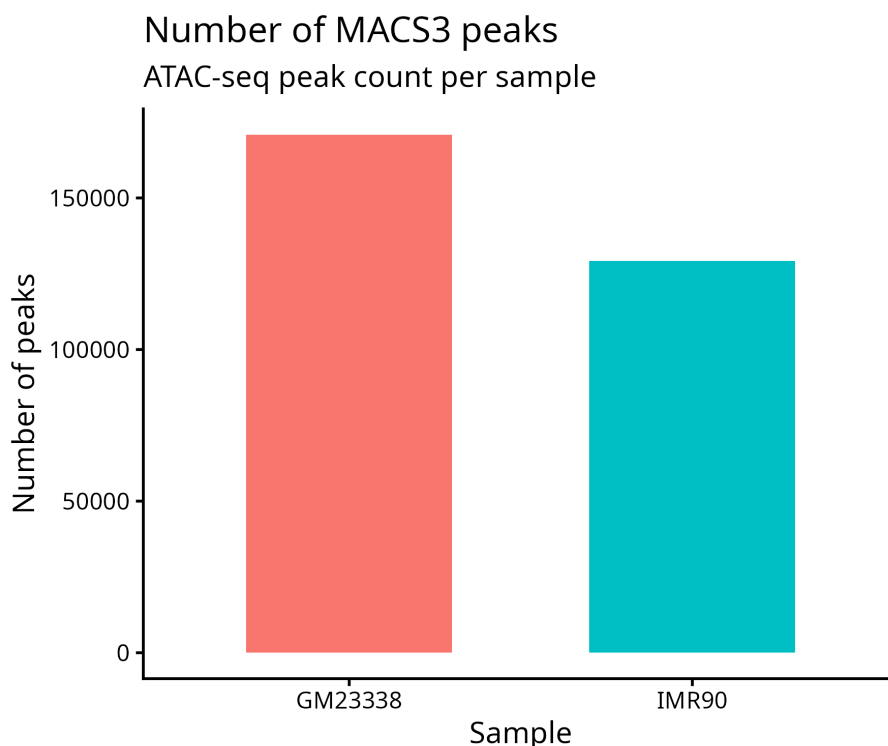
Figure 12: FRiP score- The quality checking parameter for ATAC-seq



■ ATAC-seq Peak Detection, Peak Width and Overlap

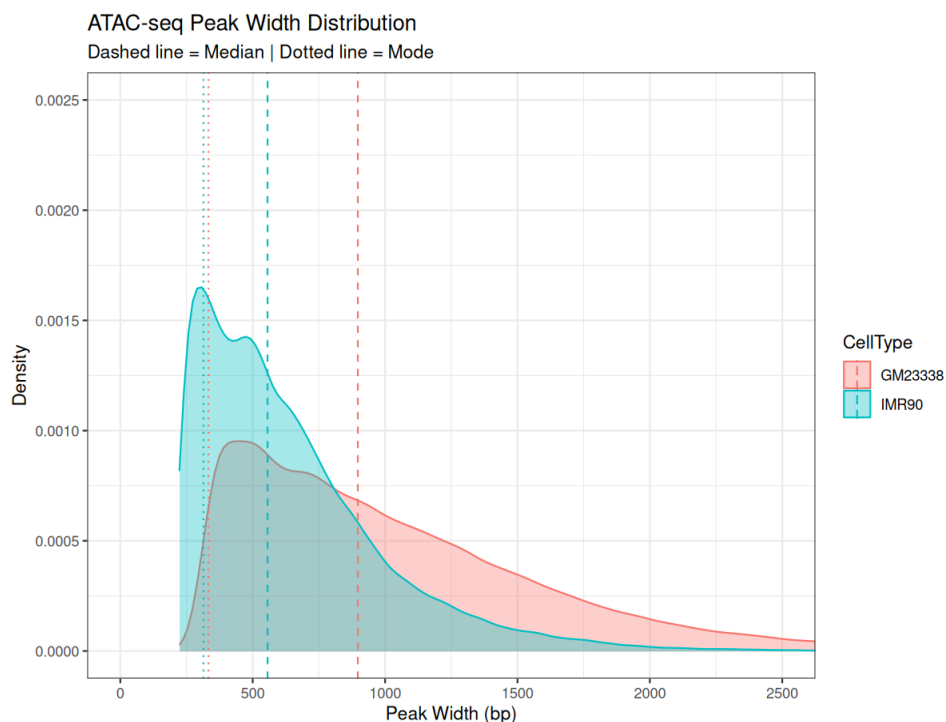
After quality check for the sequencing and the ATAC-seq, the next step would be to check the distribution of the peak counts between the different cells. Figure 13 shows that GM23338 yields ~30% more peaks overall than IMR90, showing higher baseline genome accessibility.

Figure 13: Distribution of the ATAC-seq peak counts



The width density plot (Figure 14) shows a major structural difference: IMR90 peaks are tightly bounded and narrow (~550 bp median), whereas GM23338 peaks are much broader with a heavy right tail (~900 bp median).

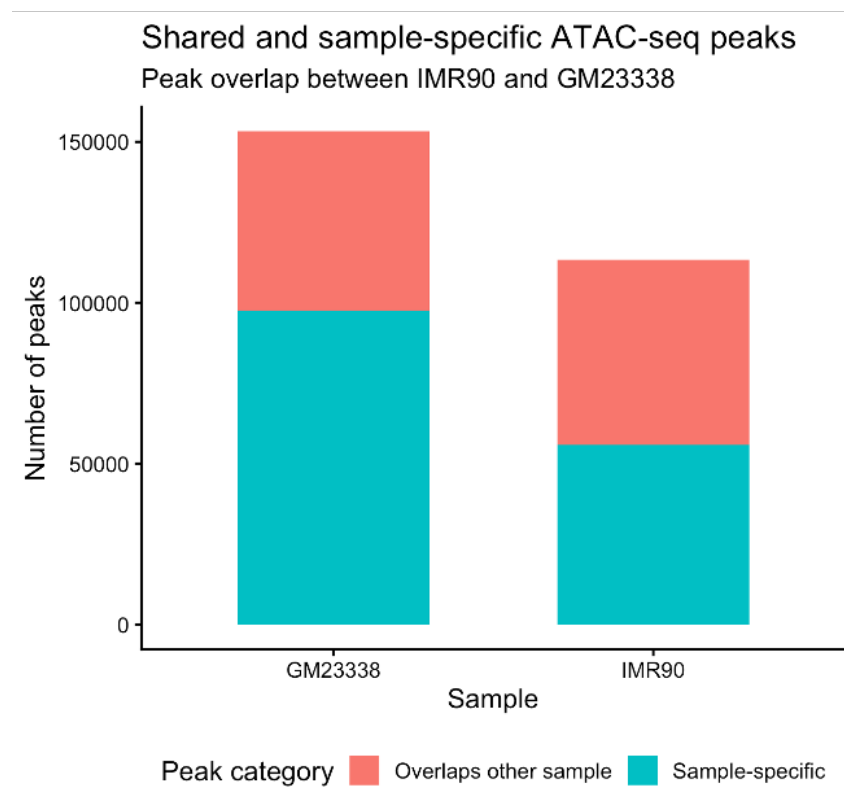
Figure 14: Distribution comparison of the ATAC-seq peak counts between cells.



Looking at set intersection (Figure 15), only about 55,000 peaks are shared (<35%), meaning the vast majority of open chromatin regions are sample-specific.

Together, these metrics demonstrate that cell identity alters both the position and the physical footprint of accessible chromatin.

Figure 15: Shared and sample specific ATAC-seq peaks



Quantitative Signal Intensity and Spatial TSS Distance of ATAC-seq Peaks

After sequencing and alignment, we have millions of reads. If many reads pile up at a particular genomic location, that suggests the DNA is accessible there. A peak therefore represents: *A genomic region showing unusually high ATAC-seq read enrichment, interpreted as an accessible chromatin region.* MACS3 can identify these peaks from the aligned BAM files.

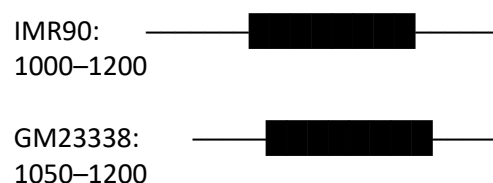
A consensus ATAC-seq peak is a genomic region that has been defined as a common, reliable accessible region across multiple ATAC-seq samples or replicates. This is especially important in our analysis because we have IMR90 and GM23338 samples and are creating a common set of peaks before comparing chromatin accessibility between them.

For example: We have two cell types

IMR90
Peak A: chr1:1000–1200
Peak B: chr1:3000–3300
Peak C: chr1:5000–5300

GM23338
Peak A: chr1:1050–1200
Peak D: chr1:7000–7300
Peak E: chr1:9000–9300

Some peaks overlap between the samples. For example:



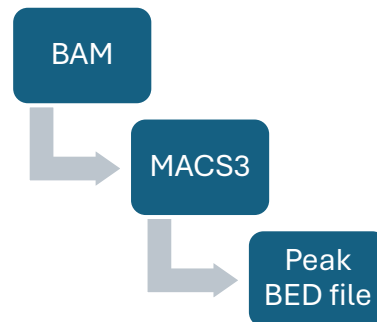
These represent essentially the same accessible genomic region. We can merge/identify these overlapping regions to create a common peak set. That common set is called a consensus peak set.

How is a consensus peak calculated?

There isn't one universal mathematical formula. A consensus peak is generally created through a **genomic interval overlap/merging procedure**. The basic process is:

Step 1- Cell peaks separately

For each sample:



For example:

IMR90_peaks.bed
GM23338_peaks.bed

Each contains genomic coordinates such as:

```
chr1 1000 1200
chr1 3000 3400
chr1 5000 5400
```

Step 2 — Combine the peak coordinates

You combine all peak regions:

IMR90:
chr1 1000 1200
chr1 3000 3400

GM23338:
chr1 1050 1250
chr1 7000 7300

Combined:

```
chr1 1000 1200
chr1 3000 3400
chr1 1050 1250
chr1 7000 7300
```

Step 3 — Identify overlapping regions

Now ask: Do these peaks overlap?

For example:

```
IMR90    1000————1200
GM23338  1050————1250
          ↑
        Overlap
```

They overlap between: 1050–1200

Therefore, they can be treated as representing the same accessible region. Depending on the pipeline, overlapping intervals may be merged, or a representative/common interval may be defined.

Consensus peak vs differential peak

These two concepts are often confused.

Consensus peak answers: *"Where are the accessible regions that we will use for comparison?"*

Differentially accessible peak answers: *"Which of those regions has significantly different accessibility between conditions?"*

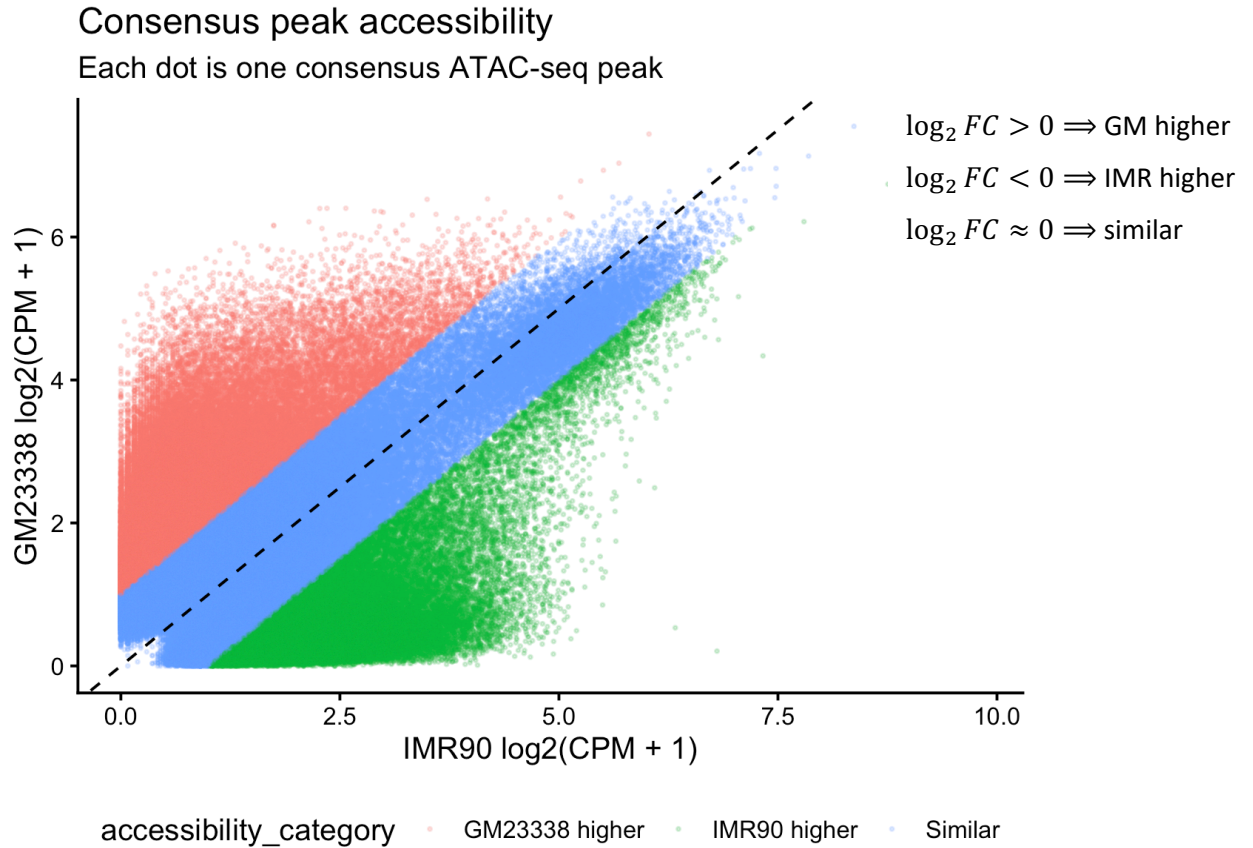
In this example we have plotted chromatin accessibility at every consensus ATAC-seq peak between IMR90 and GM23338. The key idea is: Each dot = one genomic region (one consensus ATAC-seq peak), and its position tells us how accessible that region is in the two cell types. X-axis: IMR90 accessibility IMR90 $\log_2(\text{CPM}+1)$. So, moving right

means the peak has higher ATAC-seq read abundance/accessibility in IMR90. Y-axis: GM23338 accessibility
GM23338 $\log_2(\text{CPM}+1)$ So, moving up means the peak has higher accessibility in GM23338.

$$\text{CPM} = \text{Counts Per Million} = \frac{\text{Reads in a peak}}{\text{Total mapped reads}} \times 1,000,000. \quad \log_2 FC = \log_2 \left(\frac{\text{CPM}_{\text{GM23338}}}{\text{CPM}_{\text{IMR90}}} \right)$$

CPM is a way of normalizing the number of sequencing reads/fragments assigned to a peak according to the total number of reads in that sample.

Figure 16: Consensus peak accessibility



Here, Figure 16 is the scatter plot of the consensus peaks compared between IMR90 and GM23338 cells.

The diagonal dashed line represents invariant baseline accessibility $\Rightarrow$ Common peaks between IMR90 and GM23338 $\Rightarrow$ Common chromatin accessibility area between the two cells.

Coral & Green Regions: High-contrast, cell-type-specific peaks showing marked shifts in accessibility between GM23338 and IMR90 (~51.4% combined).

Note that the $\log_2$ transformed scales were added with 1 because to avoid $\log_2(0)$ problem.

By observing the scatter plot: About 48.6% of peaks lie directly along the diagonal, representing invariant, shared baseline accessibility. However, over 124,000 peaks show strong fold-change shifts—65,828 elevated in GM23338 and 58,365 elevated in IMR90.

Figure 17: Distance of consensus peaks to the Transcription Start Site

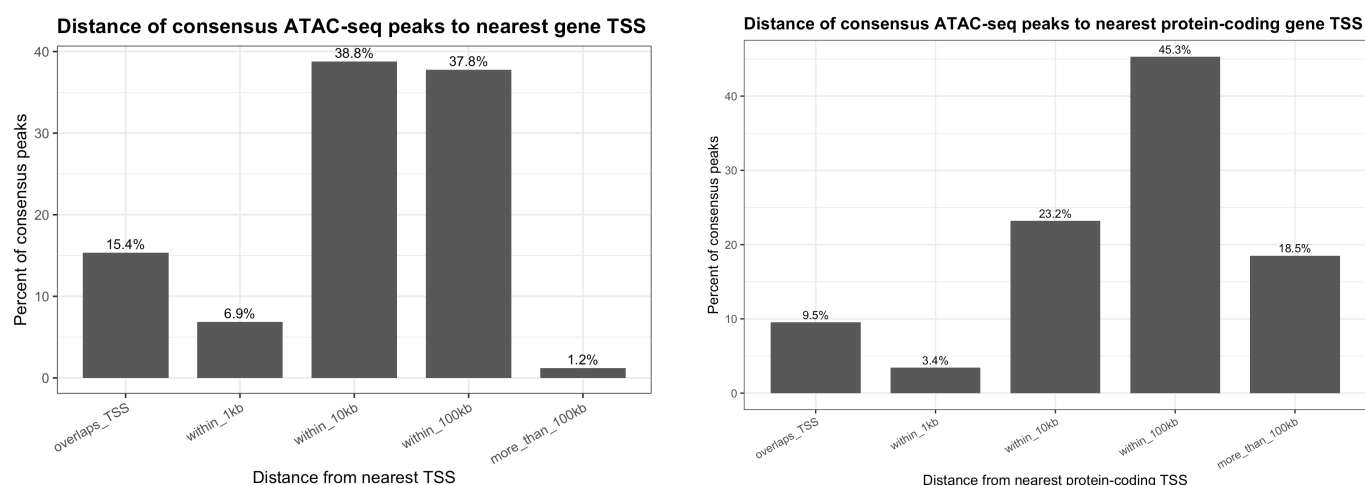
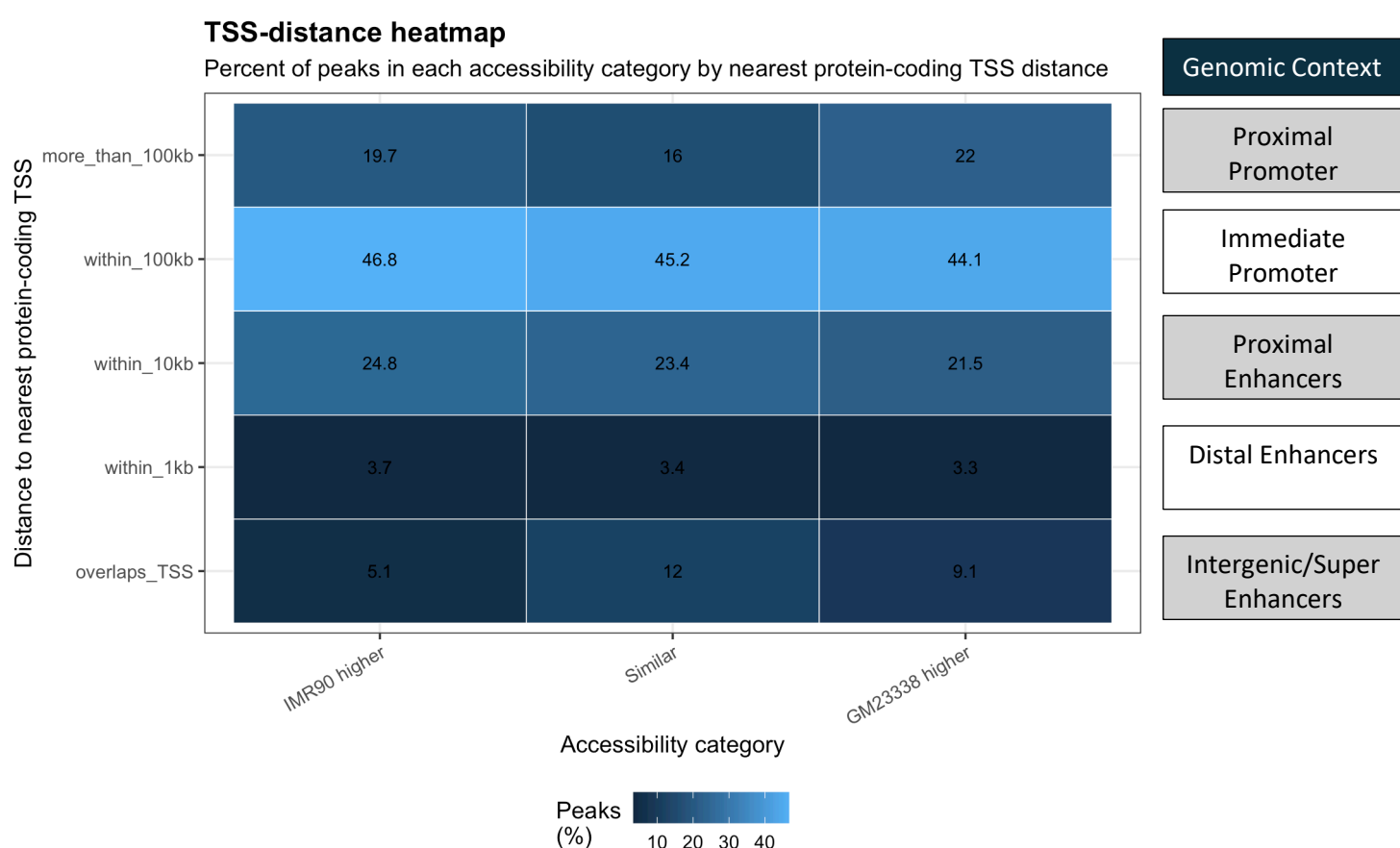


Figure 18: TSS-distance heatmap



After identification of consensus peaks between two cell types, the next step would be to identify and compare the accessibility of the genes and protein-coding genes transcription start sites (TSS) between the two cells, IMR90 and GM23338. It is essential to understand which common genes and protein-coding genes are commonly accessible between the two cells.

To achieve this, we annotate consensus ATAC-seq peaks relative to genomic features—specifically measuring their linear distance to the nearest TSS.

Step 1: Overall Genomic Annotation (All Genes vs. Protein-Coding Genes) [Figure 17]

By comparing all genomic features against strictly protein-coding genes, we can observe how accessible chromatin is distributed across non-coding vs. coding gene landscapes.

Key Analytical Insights:

- ∞ Promoter-Proximal Regions ($\leq 1\text{ kb}$):
 - When considering all genes (including non-coding RNAs, snRNAs, and pseudogenes), 22.3% (15.4% + 6.9%) of peaks fall within 1kb of a TSS.
 - When filtering strictly for protein-coding genes, promoter-proximal peaks drop to 12.9% (9.5% + 3.4%).
- ∞ Distal Regulatory Regions ($> 10\text{ kb}$):
 - Relative to protein-coding genes, 63.8% (45.3% + 18.5%) of consensus peaks reside at distal locations ($> 10\text{ kb}$ away from the TSS), with 18.5% located beyond 100 kb.
- ∞ Biological Takeaway: The majority of accessible chromatin in human cell types corresponds to distal enhancer elements, rather than promoter-proximal regions.

Step 2: **Comparing TSS Accessibility Across Differential Accessibility Categories [Figure 18]**

To determine whether promoter vs. enhancer accessibility varies by cell identity, peaks are split into three categories based on intensity: IMR90 higher, Similar (Shared), and GM23338 higher.

Key Analytical Insights:

- ∞ Notice that in all columns, over 65% to 70% of peaks lie between 10 and 100 kilobases away from a promoter. This proves that ATAC-seq accessibility is primarily capturing distal enhancers rather than just gene promoters.
- ∞ Interestingly, shared 'Similar' peaks have double the TSS overlap (12%) compared to cell-specific peaks (~5-9%). This tells us that promoters are largely invariant across cell types, while distal enhancers drive cell-type identity.

▪ Motif Enrichment Analysis

What is a Motif and Why is it Important?

A **sequence motif** is a short, conserved nucleotide sequence pattern (typically 6–20 base pairs) recognized by specific transcription factors (TFs) to regulate gene expression.

- **Biological Relevance:** Regions identified as MACS3 ATAC-seq peaks indicate open chromatin. Finding a motif inside a peak reveals *which specific TFs* are likely binding to that regulatory element.
- **Cell-Type Specificity:** Comparing motif frequencies across IMR90 and GM23338 high-contrast peaks uncovers cell-line-specific transcriptional networks (e.g., AP-1/FOS/JUN motifs driving IMR90 vs. KLF/SP motifs driving GM23338).

Motif Enrichment Analysis

Purpose

To determine if specific TF motifs are over-represented in cell-type-specific accessible regions compared to a non-differential baseline.

Workflow & Methods

1. **FASTA Extraction:**
 - Extract sequence underlying each MACS3 consensus peak using GRCh38/hg38 genomic coordinates.
2. **Motif Scanning:**
 - Scan accessible sequences against known Position Weight Matrices (PWMs) from databases like JASPAR.
3. **Contingency Table Construction:**
 - Count motif occurrence within differential peaks (Foreground) versus non-differential/opposite peaks (Background):

Group	Motif Present	Motif Absent	Total Peaks
Foreground (Sample-Higher Peaks)	a	b	a + b
Background (Comparison/Similar Peaks)	c	d	c + d

Calculating Delta Enrichment (ΔE)

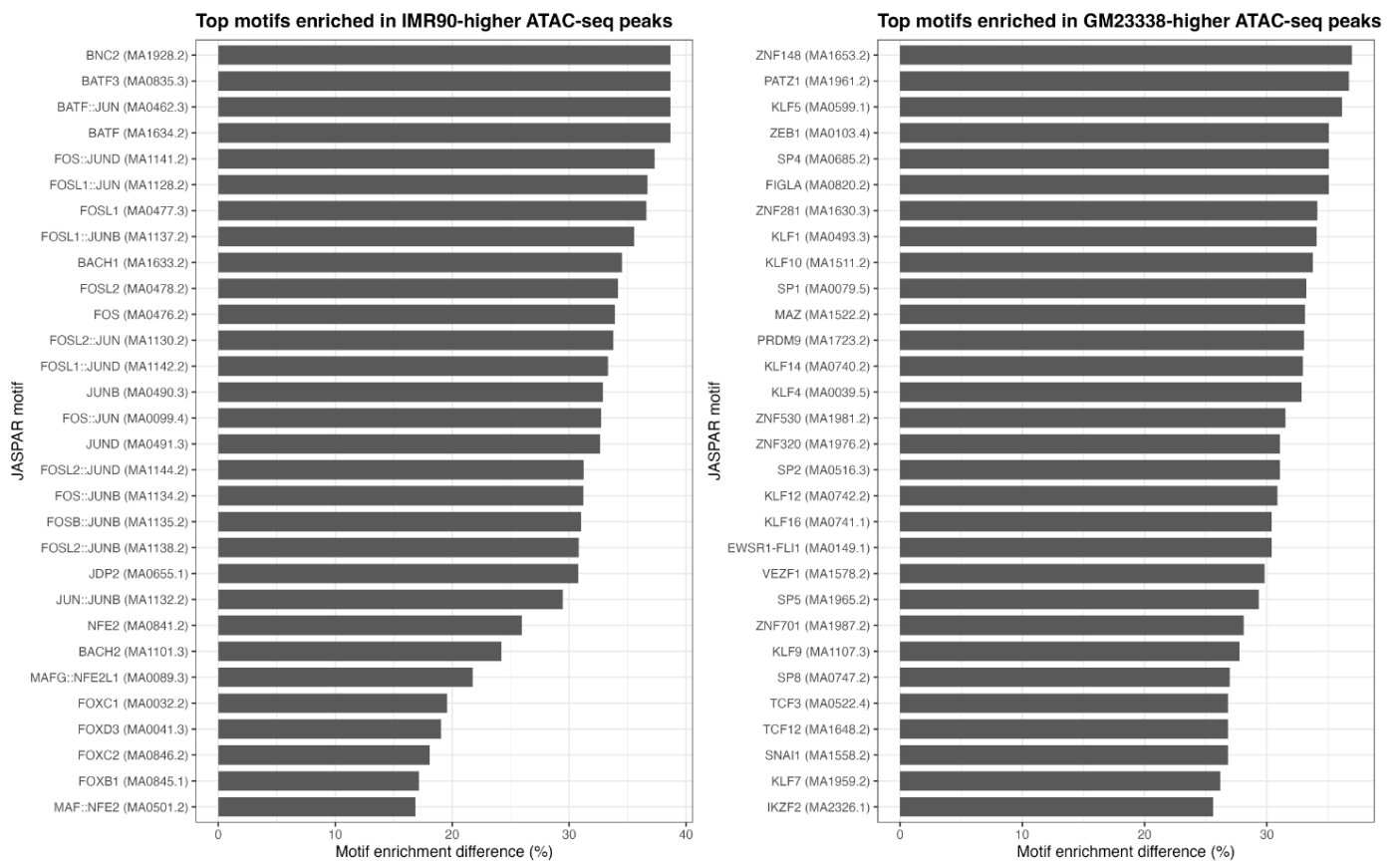
To measure the effect size ;we calculate the proportion of peaks containing motif M :

$$E_{foreground} = \left(\frac{a}{a+b} \right) \times 100$$

$$E_{background} = \left(\frac{d}{c+d} \right) \times 100$$

$$\Delta E = E_{foreground} - E_{background}$$

Figure 19: Comparison of the top motifs enriched peaks between IMR90 and GM23338



In Figure 19, we tested which transcription factor binding motifs are statistically over-represented in each cell type's unique open chromatin.

For each motif, we built a 2x2 contingency table comparing motif presence in foreground versus background peaks, calculating a Delta Enrichment percentage.

we see two distinct regulatory fingerprints:

1. IMR90 is dominated by AP-1 complex motifs like FOS and JUN, driving fibroblast identity.
2. GM23338 is enriched for Sp/KLF family motifs, characteristic of lymphoblast lineage networks.

Keep in mind, motif enrichment points to accessible regulatory 'blueprints'—it infers potential TF activity, rather than direct physical binding

We also plotted a heatmap with the top motifs. The heatmap plot JASPAR transcription factor motifs along the y-axis against the two differential MACS3 peak sets on the x-axis (**GM23338 higher peaks** vs. **IMR90 higher peaks**). Color intensity reflects the calculated **Enrichment difference (%)**: dark blue represents negative or baseline enrichment, while light blue represents high positive enrichment (~ 30%).

Figure 19: Filters strictly for the top enriched motifs per group, producing a clean two-block signature showing mutual exclusivity between cell lines.

▪ Nearest Protein-Coding Gene Mapping

The purpose to map differential ATAC-seq peaks to their nearest protein-coding genes as an exploratory, high-level screen for candidate downstream genes or pathways regulated by cell-specific enhancers/promoters.

Regulatory elements (enhancers/promoters) often modulate the nearest gene (Figure 20). This visualization counts how many high-contrast (cell-line-specific) ATAC-seq peaks lie nearest to individual protein-coding genes.

- ∞ **GM23338 higher:** Genes like *DLGAP2*, *AUTS2*, and *IGSF21* have the highest number of nearby GM-specific open peaks.
- ∞ **IMR90 higher:** Genes like *KCNMA1*, *FRMD4A*, *FMN1*, and *CDC42EP3* have the highest number of nearby IMR90-specific open peaks.
- ∞ **Caveat:** As explicitly noted on the slide, this serves as an *exploratory nearest-gene summary* and does not directly prove differential transcription/expression without accompanying RNA-seq or 3D chromatin conformation (e.g., Hi-C) data.

Figure 19: Comparison of motifs near high-contrast ATAC-seq peaks between IMR90 and GM23338

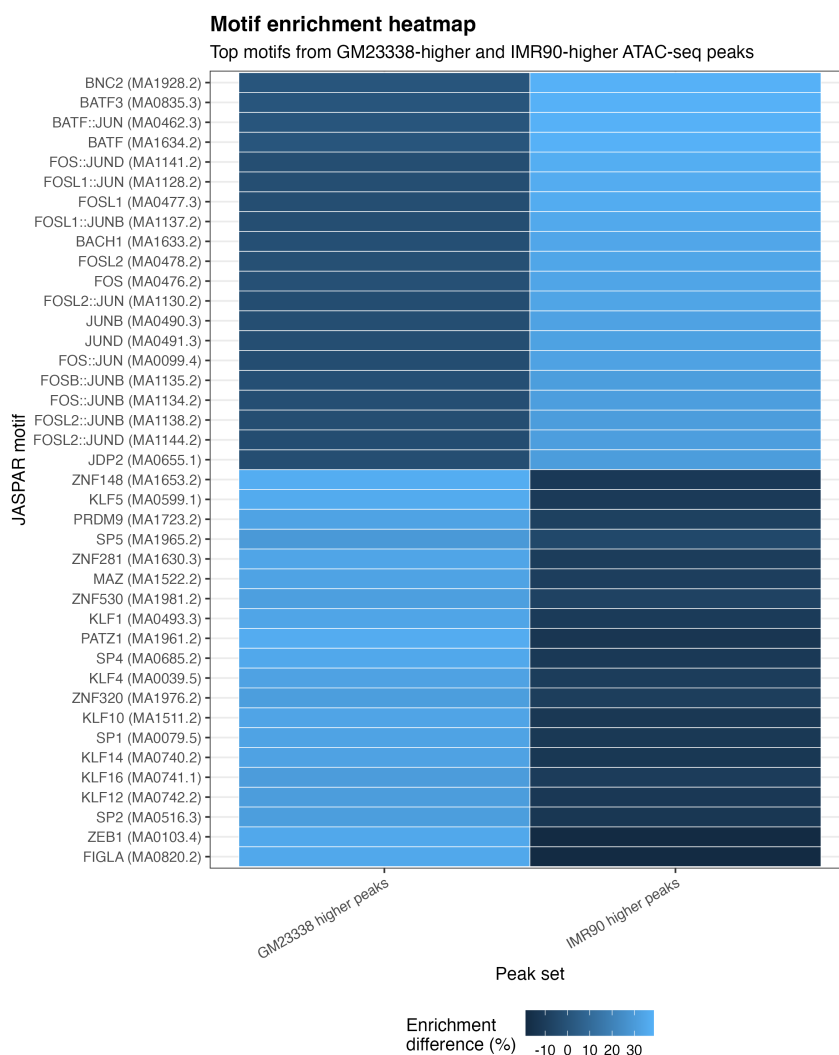
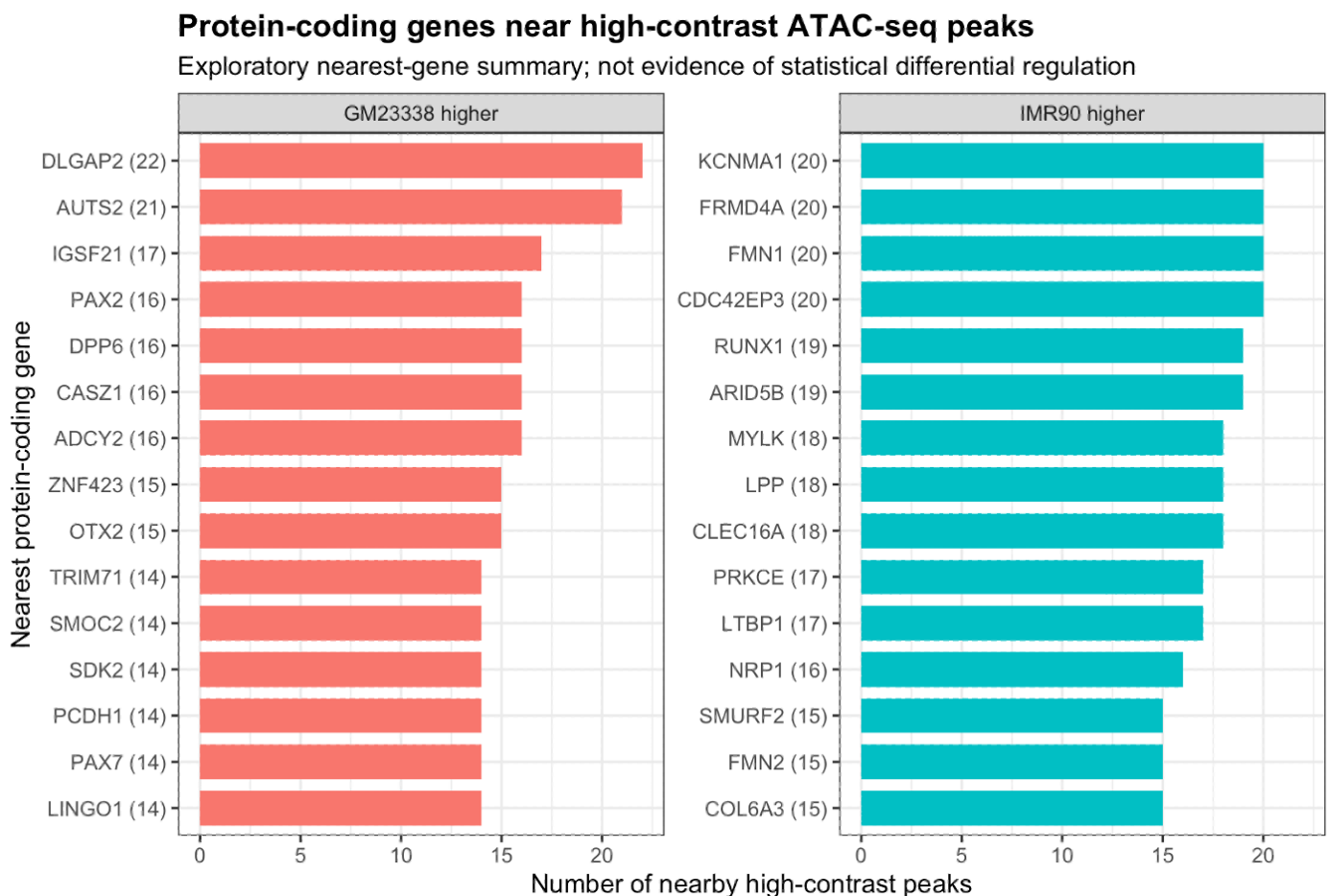


Figure 20: Comparison of protein coding genes near high-contrast ATAC-seq peaks between IMR90 and GM23338



▪ What we did here?

In this analysis, we performed a step-by-step exploratory comparison of open chromatin landscapes between two distinct human cell states: **IMR90** (a differentiated, fibroblast-like cell line) and **GM23338** (a stem-cell-like iPSC line) using bulk ATAC-seq data.

- **Quality Control & Biophysical Integrity:** Checked sequencing quality, coverage depth, and library integrity. Both samples showed high-quality sequencing with preserved nucleosomal periodicity (sub-nucleosomal, mono-nucleosomal, and di-nucleosomal peaks) and excellent Fraction of Reads in Peaks (FRiP) scores (~55% for GM23338 and ~52% for IMR90, well above the 20% minimum quality threshold).
- **Peak Detection & Genomic Footprinting:** Identified accessible regions (peaks) using MACS3. GM23338 displayed ~30% more peaks overall and broader peak widths (median ~900 bp) compared to the narrower, tightly bounded peaks in IMR90 (median ~550 bp). Less than 35% of peaks (~55,000) were shared between the two cell lines.
- **Consensus Peaks & Differential Accessibility:** Created a consensus peak set to evaluate quantitative signal intensity. While ~48.6% of regions represented shared baseline accessibility, over 124,000 peaks showed strong cell-type-specific shifts (65,828 higher in GM23338 and 58,365 higher in IMR90).
- **Distance to Transcription Start Sites (TSS):** Measured how close accessible peaks were to gene starting points. The vast majority of accessible regions (>63% relative to protein-coding genes) were located in distal regulatory regions (>10 kb away from the TSS). While shared promoter regions remained largely invariant, cell-type differences were driven predominantly by distal enhancers.
- **Motif Enrichment & Nearest-Gene Mapping:** Scanned open regions for transcription factor (TF) binding patterns and mapped high-contrast peaks to nearby genes. We identified distinct TF regulatory blueprints—AP-1 family motifs (FOS/JUN) driving IMR90 versus Sp/KLF family motifs driving GM23338—and mapped cell-specific peak clusters to nearest protein-coding genes (e.g., *KCNMA1*, *FRMD4A* for IMR90; *DLGAP2*, *AUTS2* for GM23338).

▪ Layman's Interpretation of Findings

Think of the human genome as a massive instruction manual present inside every cell. Even though a skin cell and a stem cell contain the exact same manual, they read completely different pages. ATAC-seq acts like a highlighter, revealing which pages (DNA regions) are physically open and readable, and which are tightly folded shut.

- **Stem cells keep more options open:** The stem-cell-like iPSCs (GM23338) have a much broader, more relaxed DNA structure with significantly more accessible regions than the fully specialized connective tissue cells (IMR90). Because stem cells have the potential to turn into any cell type, their DNA landscape remains wider and more open.
- **Identity is shaped by distant switches, not starting gates:** The primary "starting gates" of genes (promoters) are mostly open in both cell types. Instead, cell identity is determined by distant "light switches" (enhancers) located far away from the genes themselves. It is these far-off regulatory switches that differ dramatically between skin-like cells and stem cells.
- **Different master keys drive each cell type:** By analyzing the DNA patterns in these open regions, we found two distinct sets of molecular workers (transcription factors). Fibroblast cells rely on the **AP-1 (FOS/JUN)** team of workers to maintain structural tissue programs, whereas stem cells rely on the **Sp/KLF** team to maintain pluripotency and stem-cell characteristics.

▪ Applications of ATAC-seq

Beyond the IMR90-versus-GM23338 comparison used in this workshop, ATAC-seq has become one of the standard tools in genomics because it answers a question that sits upstream of almost every other regulatory assay: which parts of the genome are physically open for business in a given cell. This single readout supports a wide range of downstream applications.

General Uses of ATAC-seq

- ∞ **Mapping regulatory elements:** ATAC-seq pinpoints promoters, enhancers, insulators, and transcription factor footprints genome-wide, without needing a prior antibody or hypothesis about which factor is involved.
- ∞ **Defining cell identity and differentiation:** Because accessible chromatin differs by cell type, ATAC-seq can be used to track how the regulatory landscape changes as stem cells differentiate, or as normal cells transition into diseased states.
- ∞ **Interpreting non-coding disease variants:** Most variants identified by genome-wide association studies (GWAS) fall outside protein-coding genes. Overlapping these variants with ATAC-seq peaks in the right cell type helps explain how a non-coding change might disrupt gene regulation rather than protein structure.
- ∞ **Biomarker and drug-target discovery:** Regulatory regions that are selectively open in diseased tissue can point to candidate transcription factors or pathways worth targeting therapeutically.
- ∞ **Studying development and reprogramming:** As in this workshop's IMR90 (fibroblast) versus GM23338 (iPSC) comparison, ATAC-seq is widely used to study how chromatin opens or closes at pluripotency and lineage-specific genes during reprogramming or development.

Using ATAC-seq to Assess Risk of Diseases Like Cardiovascular Disease (CVD)

Genome-wide association studies have identified hundreds of loci linked to coronary artery disease and other cardiovascular traits, but the large majority of these risk variants sit in non-coding DNA. On their own, GWAS results only flag a genomic neighbourhood; they do not say which gene the variant affects, in which cell type, or by what mechanism. This is precisely the gap ATAC-seq is used to fill.

The general strategy is:

- ∞ Generate ATAC-seq (or single-cell/single-nucleus ATAC-seq) profiles from disease-relevant cell types, such as coronary artery smooth muscle cells, endothelial cells, cardiomyocytes, and macrophages/foam cells.
- ∞ Identify which genomic regions are accessible, i.e. candidate regulatory elements, in each of these cell types.
- ∞ Overlap CVD-associated GWAS variants with these accessible regions. A variant that falls inside an open chromatin peak in, say, smooth muscle cells is a strong candidate for actually altering a regulatory element in that cell type.

- ∞ Combine this with transcription factor motif analysis (as in this workshop) to ask whether the variant disrupts a known TF binding motif, which would explain why that regulatory element is affected.

Studies using this approach have shown that CVD-associated SNPs are significantly enriched in open chromatin regions of vascular and cardiac cell types, and have used ATAC-seq (including single-nucleus ATAC-seq of coronary artery tissue) to prioritise specific candidate causal variants and genes at CVD risk loci, for example implicating regulatory elements near *MRAS* and *LIPA* in coronary artery disease. In short, ATAC-seq does not diagnose CVD risk in an individual patient the way a clinical test does; rather, it is a research tool that helps translate a statistical GWAS signal into a concrete, testable biological hypothesis about which regulatory element, in which cell type, contributes to disease risk.

A Brief Look at Single-Cell ATAC-seq (scATAC-seq)

Everything described in this workshop is bulk ATAC-seq: the accessibility signal is averaged across tens of thousands of cells in a sample. This works well when a sample is a fairly pure population, but tissues such as the heart, coronary artery wall, or a tumour are mixtures of many different cell types, each with its own accessibility landscape. Averaging across a mixture can blur or completely hide cell-type-specific regulatory signals.

Single-cell ATAC-seq (scATAC-seq), and its close relative single-nucleus ATAC-seq (snATAC-seq) for frozen or hard-to-dissociate tissue, solves this by measuring chromatin accessibility in thousands of individual cells or nuclei in parallel. Each cell yields its own sparse accessibility profile, and cells are then clustered by similarity to reveal distinct cell populations directly from the data, without needing to physically sort cells beforehand. This makes it possible to ask, for example, whether a CVD risk variant sits in accessible chromatin specifically in smooth muscle cells versus macrophages within the same coronary artery sample, something bulk ATAC-seq of the whole tissue cannot resolve. The trade-off is that single-cell data are much sparser per cell than bulk data, so specialised computational tools and larger cell numbers are needed to get a reliable signal.

■ Conclusion

Chromatin accessibility is the first, and often overlooked, step in gene regulation: before a gene can be transcribed, the DNA encoding it must be physically reachable by transcription factors and RNA polymerase. ATAC-seq exploits the Tn5 transposase to mark exactly these open regions, turning an otherwise invisible latent variable, accessibility, into countable sequencing reads.

Working through the IMR90 (differentiated fibroblast-like) versus GM23338 (iPSC, stem-cell-like) comparison in this workshop illustrated the full analysis arc: from raw fragment size distributions and FRiP scores that establish library quality, through peak calling and consensus peak construction, to TSS-distance annotation, motif enrichment, and nearest-gene mapping that connect accessible regions back to biology. A consistent theme emerged throughout: most accessible chromatin lies far from promoters, in distal enhancers, and it is these enhancers, more than promoters, that differ most between cell types and that appear to drive cell identity.

More broadly, ATAC-seq's simplicity and low input requirements have made it a standard tool for mapping regulatory elements, tracking cell identity during development and reprogramming, and interpreting the non-coding genetic variants uncovered by GWAS, including variants associated with cardiovascular disease risk. Its single-cell extension, scATAC-seq, pushes this further by resolving accessibility cell type by cell type within complex tissues. Taken together, these methods are steadily filling in the regulatory layer that sits between "which genes exist" and "which genes are actually used", and why that answer differs from one cell, and one disease state, to the next.

■ Further Reading

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